

Research Article

Foundation or Formula? A Simulation-Based Comparison of Google's TimesFM-3 and Classical Time-Series Models

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Abstract

Time-series foundation models such as Google's TimesFM-3 forecast series they were never trained on. Whether they should replace classical methods such as ARIMA and exponential smoothing is difficult to settle on public benchmarks, because few benchmark datasets are absent from every model's pre-training corpus. This exploratory study describes, under controlled conditions, how TimesFM-3 behaves relative to classical methods and in which regimes it may be preferred. The evaluation data are generated from nine known processes, so the model cannot have seen the series. Across 7 200 series and a pre-registered protocol with a 12-step horizon, neither family dominates. TimesFM-3's error is never more than 1.31 times that of the best method in a scenario, whereas every classical method's is at least 2.2 times somewhere; the largest classical failures occur with only two seasonal cycles, where the automatic methods fall back to non-seasonal models, and without those scenarios automatic ARIMA's worst case is 1.62. With four or more cycles automatic ARIMA is 21–24% more accurate than TimesFM-3 on seasonal ARIMA data. Among five foundation models, this robustness is specific to TimesFM-3, and it is specific to short horizons: at steps 25 to 48, the three foundation models run at that horizon forecast a decline on a saturating process whose level stays flat, and automatic ARIMA has the smaller worst case elsewhere. On intermittent demand TimesFM-3 shows no advantage over simple benchmarks built from the history. In post-hoc checks, randomised parameters, heavy-tailed noise and outliers, alone or together, leave these patterns qualitatively unchanged, although testing for seasonality narrows the classical worst case; on the official test period of 1 000 M4 monthly series TimesFM-3 is level with the fifth- to seventh-ranked M4 entries, behind the four best, and on 101 macroeconomic series observed after the models' documented training data it is statistically level with the classical methods. Because TimesFM-3 is a black box, the study characterises its behaviour, not the reasons for it; the findings are conditional on the processes and horizons examined, and the model's weights are licensed for non-commercial use only.

1 | Introduction

- Time-series foundation models are large neural networks, pre-trained on many series and then applied to new series without task-specific training. On 31 August 2026 Google Research released TimesFM-3, a 330-million-parameter decoder-only transformer and the first member of its family designed for multivariate forecasting, which its developers report as

the top-ranked pre-trained model on three public benchmarks [1]; such claims are hard to verify, for the reason given below. For applied forecasting the relevant question is whether such a model should be used, in preference to a classical method such as ARIMA, for the series at hand; the two families also differ in cost, interpretability, the basis of their prediction intervals and the terms under which they may be used.

Published benchmarks are poorly suited to answering that question. Of 401 datasets used to evaluate 22 time-series foundation models, only about 6% had never appeared in any model's pre-training corpus, and even a 0.1% overlap can move the mean absolute percentage error by 8 to 29 percentage points [2]. Foundation-model evaluations have also been criticised for comparing against under-tuned classical baselines rather than the combinations that win forecasting competitions [3]. This study therefore generates its own evaluation data. Under a known data-generating process (DGP) the foundation model cannot have seen the series, because they did not exist before the experiment, and the model families searched by AutoARIMA and AutoETS contain the true process of the linear and exponential-smoothing DGPs. What the design excludes is contamination at the level of the realisation; what it cannot exclude is familiarity at the level of the family, because TimesFM-3 is pre-trained partly on synthetic series [1]. Such familiarity cannot be measured here, and its direction cannot be assumed: synthetic corpora contain autoregressive and seasonal structure, resembling D1–D5, but also level shifts, piecewise trends and sparse spikes, resembling D6–D8. No conclusion of this paper relies on its direction. The aim of the study is exploratory: to describe, under controlled conditions, how the accuracy of TimesFM-3 relative to classical methods varies across regimes, and to derive diagnostic baselines for when it may be preferred and when it should not. It proposes no new estimator, test or theory. The design can establish what TimesFM-3 does but not why: the 330 million parameters have no individual interpretation and the pre-training corpus cannot be inspected, so explanations of its behaviour offered here are interpretations consistent with the measurements, not findings.

The study makes four contributions. First, it evaluates a time-series foundation model on data that are free of realisation-level contamination by construction, under a protocol frozen before any forecast was produced, with every departure recorded. Second, it characterises relative accuracy by the worst case across regimes as well as by the average, and shows that, among five foundation models, a bounded worst case is specific to TimesFM-3 and confined to horizons of up to two years of monthly steps. Third, it separates correct specification from estimability: the largest failures of automatic ARIMA and ETS arise when two seasonal cycles do not suffice to fit a seasonal form. Fourth, it places the foundation models against the published forecasts of the leading M4 submissions and against simple history-based benchmarks for intermittent demand, and it records the licensing and cost constraints that bear on their use. Section 2 describes TimesFM-3, Section 3 the design and the pre-registered hypotheses, Section 4 the pre-registered comparison, Section 5 the post-hoc analyses of the simulation, Section 6 the application to M4 and to data observed after the training corpora, and Section 7 the discussion.

1.1 | Related work

TimesFM [4], Chronos [5], Moirai [6], TimeGPT [7] and Lag-Llama [8] established that a single pre-trained network can forecast unseen series zero-shot, and GIFT-Eval [9] and fev-bench [10] supplied common benchmarks that include AutoARIMA, AutoETS and seasonal naive among their baselines; newer models include MOMENT [11], TTM [12], Time-MoE [13], Sundial [14], Toto [15] and Moirai 2.0 [16]. The comparison of statistical and machine-learning methods has a longer history in this literature [17], and the critical literature on foundation models has grown alongside: Meyer, Kaltenpoth, Zalipski, and Müller [2] on contamination, and Adler, Chang, Draxler, Abdi, and Smyth [18], who found on six public datasets that the calibration error of five foundation models grows with forecast length. The present study differs from the latter in using generated rather than public data, in testing accuracy differences with multiplicity control rather than only summarising calibration, and in evaluating TimesFM-3. Applied comparisons on real data are appearing in several fields—Chu, Szeto, and Chung [19] compare six foundation models with SARIMA on influenza surveillance data—but they cannot say why a model works, because the generating mechanism is unknown.

The design follows a pattern established in this journal, a controlled simulation over several DGPs followed by an empirical application [20, 21, 22], and it checks the realism of the simulated series in a feature space, as in Kang, Hyndman, and Smith-Miles [23] and Talagala, Hyndman, and Athanasopoulos [24]. AutoARIMA and AutoETS follow the Hyndman–Khandakar procedures [25, 26], Theta [27] won M3 [28], and the combination is included because combinations are the standard against which competition entries are judged [28, 3]. Evaluation follows the scale-free conventions of Hyndman and Koehler [29] and the M5 competition [30, 31] and avoids the pitfalls catalogued by Hewamalage, Ackermann, and Bergmeir [32].

56 **2.1 | Architecture and pre-training data**

57 TimesFM-3 is the third generation of the decoder-only TimesFM family [4, 1]. Each input series is normalised by an
58 iterative reversible instance normalisation, with linear detrending of strongly trending inputs, and divided into non-
59 overlapping patches of 32 time points, each embedded as a token; 20 transformer layers with model dimension 1280
60 and 16 attention heads give about 330 million parameters (the checkpoint loaded here has 330 710 976). Layers of causal
61 attention along time alternate with layers of attention across variates, so target series and covariates can be processed
62 jointly, and a contiguous patch masking scheme produces the whole horizon in one forward pass, in output patches of 64
63 points. For each step the model returns nine quantiles, the 10th to the 90th percentile, and its point forecast is the median
64 [1, 33, 34]. These details come from the developers’ blog post and model card and from the configuration file of the check-
65 point (accessed 23 September 2026); none has yet been described in a peer-reviewed publication. The pre-training corpus
66 of more than one trillion time points combines the GIFT-Eval pre-training collection, Wikipedia page views to November
67 2023, Google Trends queries to the end of 2022, and synthetic and augmented series [33].

68 **2.2 | Checkpoint and configuration**

69 The released weights are loaded with the developers’ Python package (versions and revisions in Section 9). The
70 model is used zero-shot: it receives only the raw context, with no fine-tuning, covariates or seasonal period, through
71 `predict_batch` with its default settings (no symmetric averaging, no positivity constraint, 32-bit arithmetic). Its point
72 forecast is the model’s own median output; the nine deciles are then sorted, which matters only where they cross
73 (Section 5.2). A context shorter than the model’s working length is left-padded with zeros and masked, so at $n = 24$ the
74 model sees one partly filled patch; the package’s maximum context of 15 360 points truncates no series in this study, and
75 a series forecast alone or inside a batch of other lengths gives quantiles that agree to within 2×10^{-5} . Because no quantile
76 outside the 10th and 90th percentiles is available, the widest interval that can be formed is the nominal 80% interval.

77 **3 | Design of the simulation study**

78 Table 1 summarises the design in the aims–data–estimands–methods–performance form recommended for simulation
79 studies by Morris, White, and Crowther [35].

TABLE 1 | Structure of the simulation study.

Element	Specification
Aims	Describe how the accuracy of zero-shot TimesFM-3 relative to classical methods varies across known data-generating regimes; exploratory.
Data-generating mechanisms	Main design: nine processes (Table 2) with fixed parameters, $n \in \{24, 48, 96, 200\}$, $H = 12, 200$ replications per scenario. Post hoc: parameters drawn per series from wide ranges in five versions (Section 5.5); the nine processes regenerated for $H = 48$ (Section 5.6).
Estimands	For each method and scenario: the expected MASE over $h = 1, \dots, 12$ (and RMSSE, scaled pinball loss, 80% coverage); for each pair of methods and scenario: the pseudo-median of the paired MASE difference. Summaries across scenarios (worst ratio, win counts) were defined post hoc.
Methods	Seasonal naive, Theta, AutoETS, AutoARIMA, their combination and TimesFM-3 (pre-registered). Post hoc: DOTM, the M4 Comb benchmark, a second combination, a median combination, an Oracle ARIMA, the Croston family, the all-zero and history benchmarks on D8, TimesFM-2.5 (two settings), Chronos-Bolt, Chronos-2 and TiRex.
Performance measures	Mean MASE with its Monte Carlo standard error; ratio to the best method in a scenario, with bootstrap intervals for summaries across scenarios; paired Wilcoxon tests with Hodges–Lehmann estimates and Benjamini–Hochberg control; coverage and width of 60% and 80% intervals. With 200 replications the Monte Carlo standard error of a paired MASE difference is about 2% of the mean (Section 4.2).

Note: The mean MASE (descriptive) and the pseudo-median difference (inferential) are different estimands and can disagree when the error distribution is skewed, as on the intermittent process D8 (Section 4.2); both are reported.

3.1 | Data-generating processes

Nine processes were chosen to span the range from “a classical model is exactly right” to “no classical model is right”. Table 2 lists them. Each series is generated at length $n + H$ with $H = 12$; the first n observations form the context supplied to every method, and the final 12 are held out as the truth. One realisation of each process is shown in Figure 1.

TABLE 2 | The nine data-generating processes and their parameters.

ID	Data-generating process	Parameters	Correctly specified by
D1	AR(1)	$\phi = 0.7$	ARIMA
D2	ARMA(1,1)	$\phi = 0.6, \theta = 0.4$	ARIMA
D3	ARIMA(1,1,1) with drift	$\phi = 0.5, \theta = 0.3, \text{drift } 0.3$	ARIMA
D4	SARIMA(1,0,0)(1,1,0) ₁₂	$\phi = 0.6, \Phi = 0.5$	ARIMA
D5	ETS(A,A,A)	$\alpha = 0.3, \beta = 0.01, \gamma = 0.2$	ETS
D6	Local level with a break	break at $0.7n$, jump 10σ	neither
D7	Logistic growth	$K = 180, r = 0.08, \text{floor } 20$	neither
D8	Intermittent demand	$p = 0.3, \text{size } 1 + \text{Pois}(4)$	neither
D9	AR(1) with GARCH(1,1) errors	$\phi = 0.6, a = 0.1, b = 0.85$	ARIMA in mean

Note: The last column states which classical family is correctly specified by construction; for D6–D8 none is, and for D9 the conditional mean is correctly specified while the conditional variance is not. Innovations are Gaussian with standard deviation $\sigma = 2$ ($\sigma = 3$ in D7; D8 has none). D1, D2 and D9 fluctuate about a mean of 100 after a burn-in of 300 steps (D9: $\omega = 0.1$); D3 starts at 100 after its differences are burnt in; D4 starts from the cycle $100 + 10 \sin(2\pi k/12)$ plus noise, its seasonal differences burnt in; D5 starts from level 100, trend 0.2 and seasonal states $10 \sin(2\pi k/12)$; in D6 the level is a random walk from 100 with innovation standard deviation 0.5 and σ is that of the observation noise; D7 has lower asymptote 20; in D8 a demand occurs with probability p .

The parameter values in Table 2 were fixed in the pre-registered protocol, each chosen so that its process represents its regime unambiguously. The linear processes D1–D4 and the conditional mean of D9 use coefficients no larger than 0.7 in absolute value, well inside the stationarity and invertibility regions [36]. The smoothing parameters of D5 lie inside the usual region $0 < \beta < \alpha < 1, 0 < \gamma < 1 - \alpha$ [37], with a small trend parameter so that the level stays positive. The break in D6 equals ten noise standard deviations, so the comparison concerns adaptation to the break rather than its detection [38]; D7 follows the logistic curve, the standard model of saturating growth [39]. In D8 the mean interval between demands is $1/0.3 \approx 3.3$ periods and the squared coefficient of variation of the sizes is $4/25 = 0.16$, so against the cut-offs of Syntetos, Boylan, and Croston [40] the process is intermittent rather than lumpy, the regime for which Croston’s method was developed [41]. The GARCH(1,1) errors of D9 [42] have persistence $a + b = 0.95$, strongly persistent but covariance-stationary [43], a standard benchmark specification in volatility forecasting [44]. Every series has a level of about 100, or a positive lower asymptote in D7, so percentage errors are defined except where zeros are intrinsic (D8); the trend parameter of D5 and the asymptote of D7 were adjusted for this reason before any forecast was produced. Three design choices need a word. The break in D6 is placed inside the context, at $0.7n$: a break in the held-out window is unforecastable by any method, whereas one in the context asks whether a method forecasts from the new level or reverts to the old one. The inflection of D7 sits at $0.6(n + H)$, so the held-out window lies in the saturating arm, where a method that fits a linear drift overshoots. And the shortest length, $n = 24$, leaves a seasonal model only two cycles from which to estimate its seasonal structure, which separates whether a model is correctly *specified* from whether it is *estimable*. Two hundred replications per (DGP, length) scenario give $9 \times 4 \times 200 = 7\,200$ series, each seeded from its own identifiers.

3.2 | Forecasting methods

No method is tuned by hand; each classical model uses its own automatic selection, the comparison a practitioner faces. The classical models are fitted with StatsForecast [45] at its default settings, stated because the results depend on them. AutoARIMA searches stepwise over $p, q \leq 5$ and $P, Q \leq 2$ with $p + q + P + Q \leq 5$, chooses $d \leq 2$ and $D \leq 1$ by unit-root tests, allows a mean or drift and selects by AICc; AutoETS searches the full error–trend–season space (ZZZ) with damped and undamped trends by AICc; Theta is the standard theta method with classical multiplicative seasonal adjustment [46]. The equal-weight combination of Theta, AutoETS and AutoARIMA is included because the classical state of the art is a combination [3, 47]. Its point forecast is the mean of the members’ forecasts and its quantiles the decile-by-decile mean of theirs (vincentization), which gives a narrower distribution than a mixture when the members disagree. Every classical method receives the true seasonal period ($m = 12$ on D4 and D5, $m = 1$ elsewhere), whereas TimesFM-3 receives only the

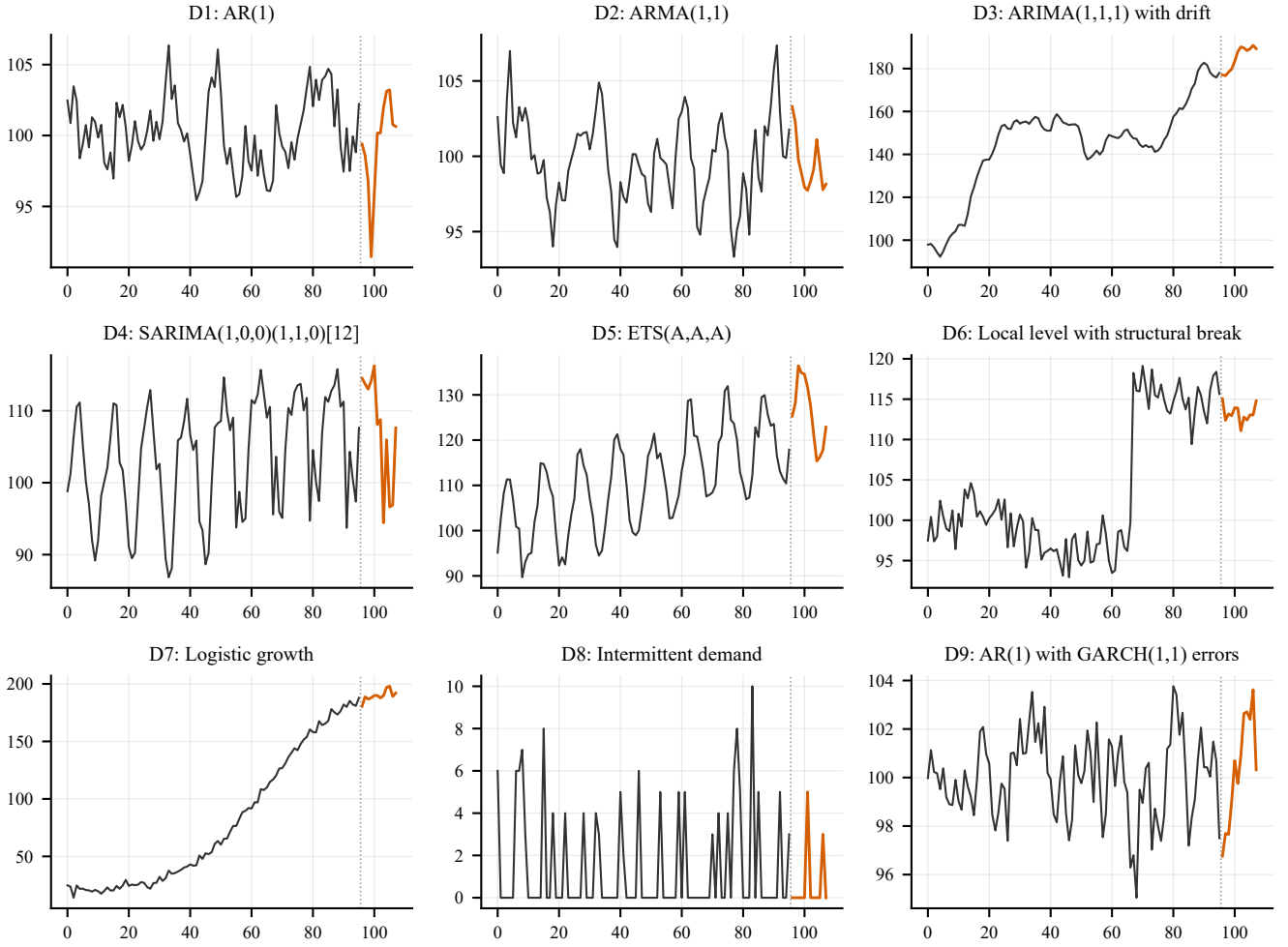


FIGURE 1 | One realisation of each data-generating process at $n = 96$. The context supplied to every method is in grey and the held-out horizon in colour, separated by the dotted line.

raw context; this reflects ordinary practice—an analyst usually knows that data are monthly—and Section 5.5 examines the alternative of testing for seasonality.

3.3 | Evaluation measures and statistical inference

Point accuracy is measured mainly by the mean absolute scaled error [MASE; 29]. With context $y_1, \dots, y_T$ ($T = n$), horizon H and point forecasts $\hat{y}_{T+h}$,

$$\text{MASE} = \frac{\frac{1}{H} \sum_{h=1}^H |y_{T+h} - \hat{y}_{T+h}|}{\frac{1}{T-m} \sum_{t=m+1}^T |y_t - y_{t-m}|}, \quad (1)$$

so the scale is the in-sample seasonal-naive error for the seasonal processes and the naive error for the others ($m = 12$ throughout for the M4 tier, as in M4). The root mean squared scaled error (RMSSE) replaces absolute by squared errors in both numerator and denominator and takes the square root, and sMAPE is the symmetric percentage error of the M4 competition. MAPE is undefined when an actual value is zero, which happens only on D8, and is reported only in the Supporting Information (Section S1) [29, 48]. With $m = 12$ and $n = 24$ the MASE denominator rests on 12 seasonal differences: its coefficient of variation across replications is 0.33 on D4 (0.09 at $n = 200$) and 0.34 to 0.44 on D5 at every length, so absolute MASE levels in those scenarios are imprecise, whereas the paired ratios used for inference are much less affected. Probabilistic accuracy is measured by a strictly proper scoring rule [49], the pinball loss over the nine deciles

125 $\tau \in \{0.1, \dots, 0.9\}$ scaled by the same denominator,

$$\text{SPL} = \frac{\frac{1}{9H} \sum_{\tau} \sum_{h=1}^H \max\{\tau(y_{T+h} - \hat{q}_{T+h}^{(\tau)}), (1 - \tau)(\hat{q}_{T+h}^{(\tau)} - y_{T+h})\}}{\frac{1}{T-m} \sum_{t=m+1}^T |y_t - y_{t-m}|}, \quad (2)$$

126 where $\hat{q}_{T+h}^{(\tau)}$ is the τ -quantile forecast, together with the coverage and width of the nominal 60% and 80% intervals [50].
127 Wider intervals cannot be formed from TimesFM-3's nine deciles, so $[q_{0.1}, q_{0.9}]$ is the widest interval compared; classical
128 quantiles are taken from intervals at levels 20, 40, 60 and 80, which map onto the same deciles.

129 Within each (DGP, length) scenario and horizon slice ($h = 1, 1-6$ and $1-12$), TimesFM-3 is compared with each classical
130 method by a paired Wilcoxon signed-rank test [51] on per-replication MASE differences, with the Hodges–Lehmann
131 estimate [52] and the median MASE ratio as effect sizes, and the Benjamini–Hochberg procedure at a false discovery rate
132 of 0.05 within each opponent family of 108 tests (36 scenarios and three slices), as pre-registered [53]; the results report
133 the $h = 1-12$ slice. The Diebold–Mariano test [54] is not used: the replications are independent, so there is no serial
134 dependence for it to model, and in the real-data tier three rolling origins per series leave it without power even with
135 small-sample corrections [55]. The pre-registered protocol had reserved it for that tier; the departure is recorded.

136 3.4 | Protocol and pre-registered hypotheses

137 The protocol was written and frozen on 8 September 2026, before any forecast was produced; it fixed the processes, lengths,
138 methods, measures and tests above and five hypotheses. **H1**: on D1–D5, where a classical family is correctly specified,
139 AutoARIMA or AutoETS is more accurate than TimesFM-3, with the gap largest at $n = 24$. **H2**: on D6–D8, where none is,
140 TimesFM-3 is more accurate than every classical method. **H3**: on D9 the two families are indistinguishable in the mean,
141 but the classical intervals under-cover. **H4**: the combination beats every individual classical method and is the hardest
142 opponent for TimesFM-3. **H5**: the 80% intervals of both families under-cover, TimesFM-3's more at longer horizons. The
143 protocol stated no decision rules, so the verdicts in Section 4.6 are judgements on the pattern of the pre-registered tests.
144 Everything in Section 5 and Section 6, except the rolling-origin evaluation of Section 6.2, was added afterwards and is
145 logged as a departure.

146 4 | Pre-registered comparison

147 This section reports the forecasts and tests fixed by the protocol. The summaries across scenarios used below—the worst
148 ratio to the best method of a scenario and the count of scenarios won—were defined after the protocol and are descriptive
149 summaries of the pre-registered forecasts. Figure 2 shows TimesFM-3 relative to the best classical method in each of the
150 36 (process, length) scenarios; the mean MASE of every method in every scenario, with Monte Carlo standard errors, is
151 in the Supporting Information (Table S2).

152 4.1 | Overall accuracy and pairwise comparisons

153 TimesFM-3 has the lowest mean MASE in 16 of the 36 scenarios and the classical methods in 20: AutoARIMA 10, seasonal
154 naive 4, and two each for Theta, AutoETS and the combination. Against the best classical method of each scenario, 18
155 differences survive Benjamini–Hochberg correction, 7 favouring TimesFM-3 and 11 the classical method; because that
156 comparator is selected on the same replications, these p -values are descriptive only. The inferential claims rest on the 180
157 comparisons with a fixed opponent (Figure 3): 132 are significant, 113 favouring TimesFM-3; pooling the tests into a single
158 family changes these counts by at most two (Supporting Information, Section S6). **AutoARIMA is the only classical**
159 **method that TimesFM-3 does not clearly beat** (11 comparisons won, 9 lost): 16 of the 36 intervals against it straddle
160 1, and AutoARIMA's wins are mainly on the seasonal processes from $n = 48$, whereas against the other methods, the
161 combination included, most intervals lie wholly on the side of TimesFM-3.

162 4.2 | Effect sizes and sensitivity to the choice of test

163 The median Hodges–Lehmann shifts in MASE across the 36 scenarios are -0.010 against AutoARIMA, -0.091 against
164 the combination, -0.133 against seasonal naive, -0.183 against AutoETS and -0.227 against Theta (negative favours
165 TimesFM-3). The Monte Carlo standard error of the paired difference between TimesFM-3 and AutoARIMA is 0.8% to

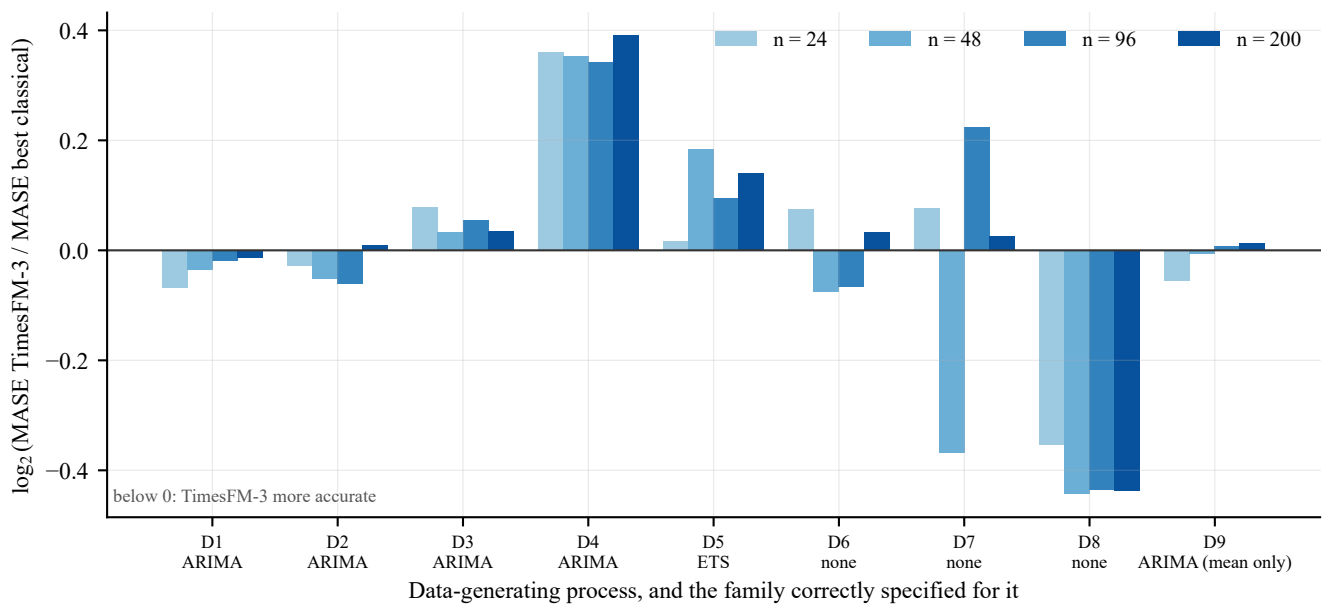


FIGURE 2 | Accuracy of TimesFM-3 relative to the best classical method in each scenario, on a $\log_2$ scale. Bars below zero mark scenarios where TimesFM-3 is more accurate. The label under each process names the family correctly specified for it by construction.

4.8% of AutoARIMA's mean MASE (median 2.1%), so differences smaller than about 6% are unlikely to be detected in a scenario: the split against AutoARIMA sits at the resolution of the design and does not show that the two methods are equivalent. A paired t -test agrees with the Wilcoxon test in 163 of 180 comparisons (90.6%). The largest block of disagreements is on the intermittent process D8 against seasonal naive: the t -test finds TimesFM-3 better at every length ($p < 0.0001$), the signed-rank test finds no difference ($p = 0.11$ to 0.92). The mean difference is driven by a minority of series on which seasonal naive fails badly, whereas on 64.5–72.0% of series TimesFM-3 is marginally less accurate—the two tests estimate different quantities of a heavily skewed error distribution.

4.3 | Correct specification versus estimability

On D1 and D2 TimesFM-3 is at least as accurate as the correctly specified AutoARIMA at every length except D2 at $n = 200$, the only significant difference: correct specification bought AutoARIMA nothing detectable on short and moderate series. The seasonal processes show why most clearly. With two complete cycles ($n = 24$), AutoETS scores a mean MASE of **3.03 on D5, data an ETS process generated**, against 1.14 for seasonal naive and 1.15 for TimesFM-3: the correctly specified model has 2.7 times the error of the naive rule. The failure is not an over-fitted seasonal model but the absence of one: given $m = 12$ and 24 observations, AutoETS chose a seasonal form for 8.5% of the D4 series and none of the D5 series, and AutoARIMA for none (Supporting Information, Table S11), whereas seasonal naive applies the known period without estimating anything. The operative distinction is therefore whether a model is *estimable* from the series at hand, not only whether it is correctly specified.

Table 3 quantifies the resulting asymmetry with each method's mean MASE as a multiple of the best in the same scenario—a relative measure, because D3 is hard for every method and would dominate absolute worst cases. **TimesFM-3 is never worse than 1.31× the best method of a scenario**, and within 0.8% of it in the median scenario, whereas every classical method is at least 2.2× worse somewhere: AutoARIMA 2.22× and AutoETS 3.50× (both on D4 at $n = 24$), Theta 4.69×, seasonal naive 5.65×. Leaving out the two scenarios with two seasonal cycles, TimesFM-3's worst ratio is unchanged and AutoARIMA's falls to 1.62 (on D7 at $n = 96$). The worst ratio depends on the methods compared, notably on seasonal naive with the known period (Section 5.5).

4.4 | Seasonal and intermittent regimes

Strong seasonality with sufficient data (D4). At $n = 48, 96$ and 200 AutoARIMA's mean MASE is 21.8%, 21.1% and 23.7% below TimesFM-3's (adjusted $p < 0.0001$), and D5 shows the same more mildly (4% to 12%). Once at least four cycles are observed, AutoARIMA recovers the seasonal structure and TimesFM-3 does not match it; seasonal naive, which

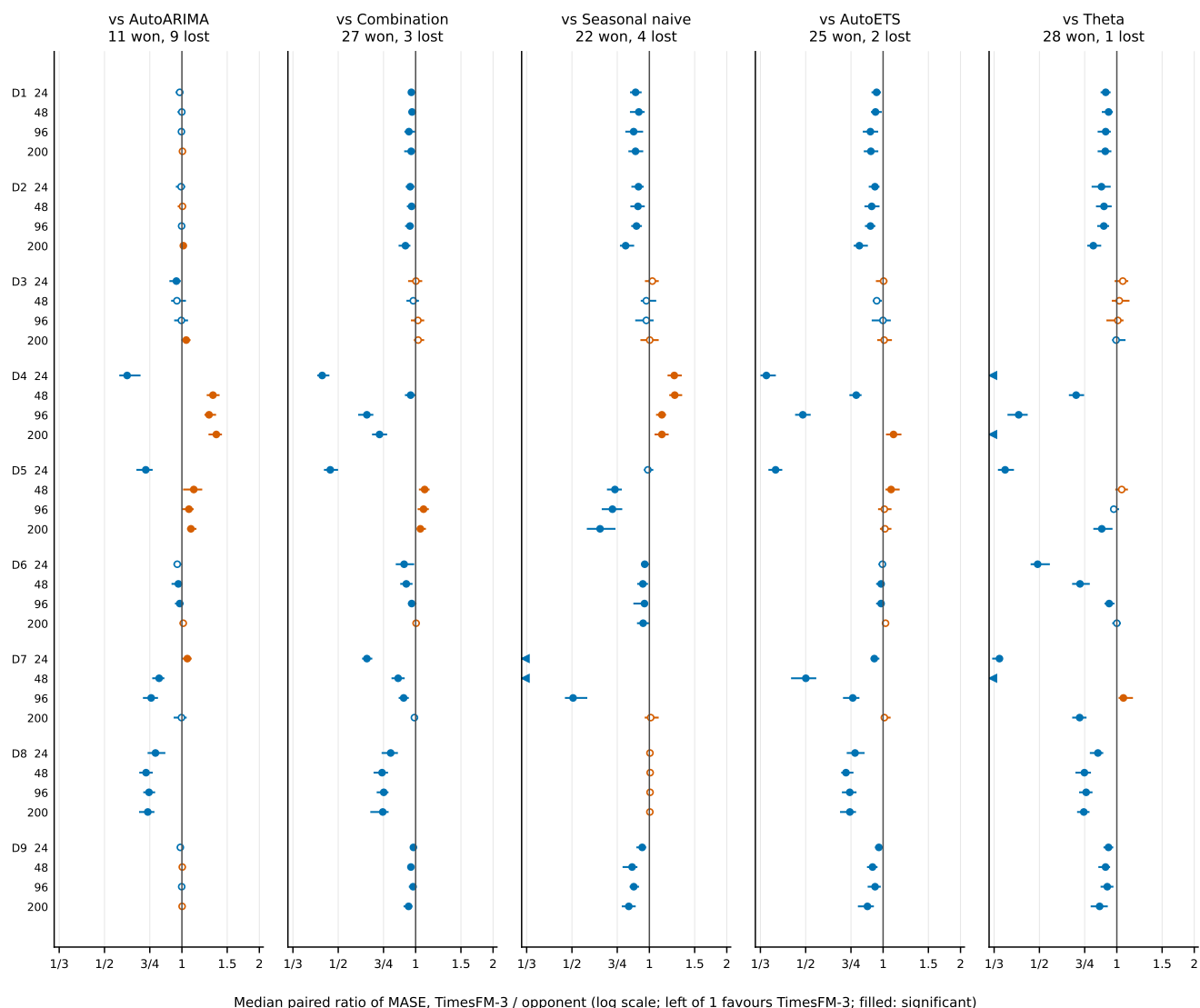


FIGURE 3 | Pairwise comparisons of TimesFM-3 with each classical method, scenario by scenario: median of the paired per-series ratios of MASE over $h = 1, \dots, 12$, with 95% bootstrap intervals, on a logarithmic scale. Filled markers are significant after Benjamini–Hochberg correction within each opponent family; points left of 1 favour TimesFM-3. Panel titles give the significant comparisons won and lost by TimesFM-3. Arrowheads mark medians beyond the plotted range.

TABLE 3 | Mean MASE relative to the best method in the same scenario, summarised over the 36 scenarios.

Method	Median	Mean	All 36 scenarios		Without the two-cycle scenarios	
			Worst	Scenario	Worst	Scenario
Seasonal naive	1.200	1.470	5.65	D7 $n = 48$	5.65	D7 $n = 48$
Theta	1.251	1.614	4.69	D4 $n = 200$	4.69	D4 $n = 200$
AutoETS	1.194	1.366	3.50	D4 $n = 24$	2.72	D4 $n = 96$
AutoARIMA	1.032	1.137	2.22	D4 $n = 24$	1.62	D7 $n = 96$
Combination	1.108	1.273	3.00	D4 $n = 24$	2.08	D4 $n = 96$
TimesFM-3	1.008	1.054	1.31	D4 $n = 200$	1.31	D4 $n = 200$

Note: Each entry divides a method’s mean MASE by the lowest mean MASE in the same (process, length) scenario, so a value of 1.00 means that the method was the best in that scenario; the mean is arithmetic. The two-cycle scenarios are D4 and D5 at $n = 24$. Geometric means and 90th percentiles are in the Supporting Information, Table S9.

estimates nothing, is also more accurate than TimesFM-3 on D4 at every length (1.005 to 1.053 against 1.124 to 1.321). With two cycles the ranking among the others reverses: TimesFM-3 beats AutoARIMA by 42% (1.321 against 2.284). Theta deteriorates on D4 as the series lengthens (1.94 at $n = 48$, 4.16 at $n = 200$): D4 is seasonally integrated, so its seasonal pattern drifts, whereas classical decomposition estimates one fixed pattern from the whole history.

Intermittent demand (D8). On MASE TimesFM-3 reduces error against the best general-purpose method by 21.7% to 26.5% at the four lengths (adjusted $p < 0.0001$) and has the lowest pinball loss of the pre-registered methods, but it *loses* on RMSSE at every length (Supporting Information, Table S6); sMAPE, which scores a zero forecast of a zero outcome as perfect, is not informative here. About 70% of D8's observations are zero: MASE is minimised by the conditional median, which is zero or near it, RMSSE by the conditional mean, which is positive, so a median-like forecast wins MASE and loses RMSSE [48, 56]. Section 5.4 shows that both advantages are matched by simple benchmarks built from the history. Across horizons ($h = 1, 1-6$ and $1-12$) every method degrades and no regime-average ranking reverses (Supporting Information, Figure S1).

4.5 | Coverage of intervals and the combination

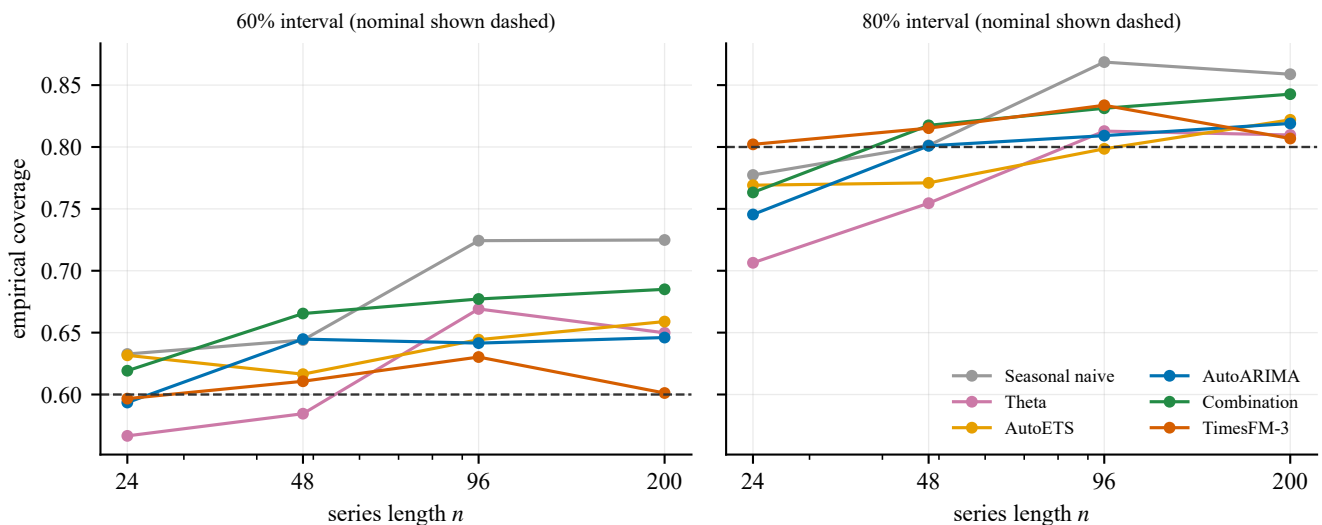


FIGURE 4 | Empirical coverage of the nominal 60% and 80% intervals, averaged over all nine processes, against series length. The nominal level is dashed. TimesFM-3's 80% coverage stays within 0.802–0.834 across every length; the classical methods under-cover at $n = 24$ and reach or exceed nominal from $n = 96$.

Averaged over the processes (Figure 4), TimesFM-3's 80% coverage stays between 0.802 and 0.834 at every length, with the smallest mean absolute deviation from nominal across lengths of any method, and within the 12-step horizon it does not fall with the horizon (0.786 at $h = 1$, 0.814 over $h = 1-12$). Averaging lets over- and under-coverage cancel, however: the mean per-scenario absolute deviation from nominal is 0.047 for TimesFM-3, 0.066 for AutoARIMA, 0.089 for the combination, 0.101 for AutoETS, 0.127 for Theta and 0.144 for seasonal naive, and TimesFM-3's per-scenario 80% coverage ranges from 0.583 to 0.921. No method delivers reliable 80% intervals on all nine processes. The classical intervals rest on Gaussian innovations, violated by construction on D8 and D9. An interval score, which rewards narrow intervals and penalises outcomes outside them (80% interval, scaled as MSIS), puts TimesFM-3 first in 22 of the 36 scenarios (Supporting Information, Table S12). On the structural break D6 every classical method covers 0.966 to 0.999, because the break inflates the estimated variance and the intervals become very wide (scaled widths 4.8 to 10.4); TimesFM-3 covers 0.828 to 0.889 with intervals about half as wide (3.0 to 5.1) and has the lowest interval score at every length. Against the pre-registered mean combination TimesFM-3 wins 27 of 30 significant comparisons.

4.6 | Assessment of the pre-registered hypotheses

Table 4 sets the pre-registered expectations against the evidence. **None of the five survives unqualified:** H1 and H5 are not supported, H4 is contradicted, and H2 and H3 are partially supported.

TABLE 4 | Pre-registered hypotheses and the evidence of the pre-registered comparison.

	Expectation	Verdict	Evidence
H1	Correct classical models win on D1–D5, most at $n = 24$	Not supported	At $n = 24$ the gap runs the other way; the classical side wins mainly where the model is also estimable (D4, D5 at $n \geq 48$)
H2	TimesFM-3 wins on D6–D8	Partially	Mildly on breaks (D6); depends on length on D7; on D8 only against the classical methods (Section 5.4)
H3	D9: tie in the mean, classical intervals under-cover	Partially	Tie in the mean; only AutoARIMA under-covers
H4	The combination beats every classical method and is the hardest opponent	Contradicted	AutoARIMA has the lower mean MASE and is the harder opponent (11–9 against 27–3)
H5	Both families under-cover, TimesFM-3 more at longer horizons	Not supported	TimesFM-3 is close to nominal on average and does not worsen within 12 steps; the classical methods reach or slightly exceed nominal from $n = 96$

Note: The protocol stated no decision rules; verdicts are judgements on the pattern of the pre-registered tests at $h = 1, \dots, 12$.

5 | Post-hoc analyses of the simulation

Everything in this section was designed after the results of Section 4 were known and is logged as a departure from the protocol.

5.1 | Precision of the summaries

The Monte Carlo standard error of a scenario's mean MASE is 1.5% to 6.2% of the mean (median 3.2%; Supporting Information, Table S2). Bootstrapping the replications within each scenario (2 000 draws, paired across methods), the worst ratio of TimesFM-3 to the scenario-best method is 1.31 (95% interval 1.28 to 1.36) and that of AutoARIMA, the smallest among the classical methods, 2.22 (2.07 to 2.38); their difference has an interval of 0.75 to 1.05. Because the best method of a scenario is chosen on the same replications, the statistic could be biased upwards; choosing it on one half of the replications and estimating the ratio on the other gives 1.32, so the bias is small. These intervals reflect Monte Carlo error only, not the choice of processes and parameters, which Section 5.5 varies. The count of scenarios won is less stable—TimesFM-3's 16 has an interval of 13 to 20—and no conclusion rests on it. Simultaneous bootstrap bands over the 36 scenarios (max- t), which control the error of the whole set of intervals, are about 1.9 times wider than the per-scenario intervals of Figure 3; against AutoARIMA they separate 8 scenarios in favour of TimesFM-3 and 4 against it, and against every other opponent they keep the direction of the pattern (Supporting Information, Table S17).

5.2 | Implementation and configuration checks

Order selection. An *Oracle ARIMA*, which knows the true orders and estimates only the parameters by Gaussian maximum likelihood, beats TimesFM-3 in 15 of the 16 ARIMA scenarios (for example 1.351 against 1.417 on D1 at $n = 24$), and on D1–D3 at $n = 24$ AutoARIMA pays a 9.9% to 13.9% penalty for selecting the orders; the short-sample deficit of the automatic method is thus largely one of model selection. On D4 at $n = 24$ most oracle fits did not converge (156 of 200, used as returned), and that scenario is not interpreted.

Implementation. The D4 and D5 scenarios were refitted with `auto.arima`, `ets` and `thetaf` of the R package `forecast` [25], the reference implementation of the Hyndman–Khandakar procedures (Supporting Information, Table S13). R reproduces every failure discussed above—AutoARIMA 2.87 and AutoETS 3.95 on D4 at $n = 24$, AutoETS 3.30 on D5 at $n = 24$, Theta 4.09 on D4 at $n = 200$ —and, like StatsForecast, almost never fits a seasonal model with two cycles. With additive instead of multiplicative decomposition Theta fails the same way on D4 (3.83 against 3.95 at $n = 200$ in 60 replications). Where both implementations choose seasonal forms they agree closely, with one exception: on D4 at $n = 96$ StatsForecast's AutoETS has a mean MASE of 2.41 against 1.04 for R, because its optimiser settles on a near-zero seasonal smoothing parameter where R finds a high one. That scenario is an implementation failure; with R's value there, AutoETS's worst ratio outside the two-cycle scenarios would be 2.13, on D7 at $n = 48$.

Combination rule. Because the mean combination is not robust to a failing member such as Theta, a *median* combination of the same three methods was tried, in the spirit of Petropoulos and Svetunkov [57]; it is slightly worse (mean MASE 1.712 against 1.656, TimesFM-3 1.439), and TimesFM-3 still wins 25 of 29 significant comparisons. Under absolute error

the mean combination can never be worse than the average of its members, by convexity, a guarantee the median does not carry.

TimesFM-3's configuration and output. The developers' benchmark evaluator averages the forecasts of a series and its negation and constrains non-negative series to non-negative forecasts. With these settings TimesFM-3's worst ratio falls from 1.31 to 1.27; no conclusion changes. The output is a 12×9 array of deciles whose raw median equals the point forecast exactly; quantile crossing occurs in 32% to 46% of steps on D8 and essentially never elsewhere. Given the month of the year as past-and-future covariates ($\sin 2\pi t/12$, $\cos 2\pi t/12$) on D4 and D5—the information the classical methods receive through their period—TimesFM-3's mean MASE rises in seven of the eight scenarios (for example 1.124 to 1.194 on D4 at $n = 96$), so supplying the period in this form did not close AutoARIMA's seasonal advantage. Using the mean of the deciles instead of the median as point forecast changes no process by more than 0.01 except D6 (0.835 to 0.875) and D8.

5.3 | Stronger classical benchmarks and other foundation models

Nine configurations were added to the six pre-registered methods (Figure 5; Supporting Information, Table S10): dynamic optimised Theta [DOTM; 58]; the Comb benchmark of M4, the mean of simple, Holt and damped exponential smoothing on seasonally adjusted data [3]; a combination of AutoETS, AutoARIMA and DOTM; TimesFM-3 with its evaluator settings; TimesFM-2.5 [59], run with its evaluator-style settings and, to match the main TimesFM-3 runs, with flip invariance and positivity switched off; Chronos-Bolt (Base, 205 million parameters) [5, 60]; and Chronos-2 [61] and TiRex [62], with 120 and 35 million parameters. All foundation models were run zero-shot with the raw context only.

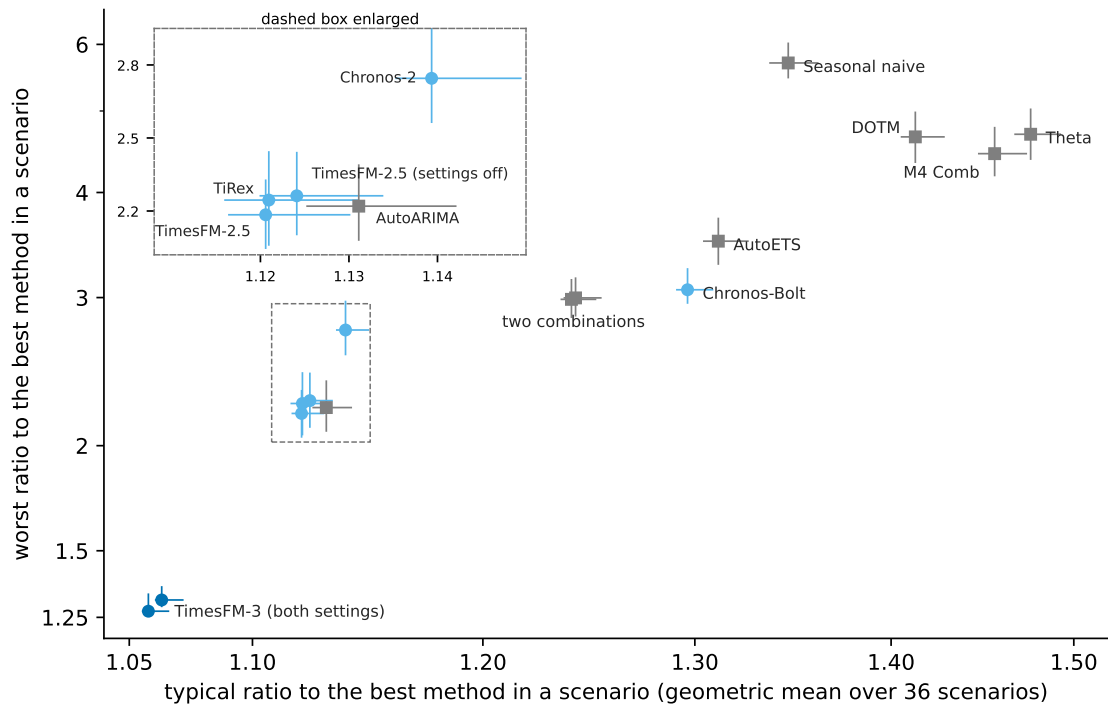


FIGURE 5 | Typical against worst-case accuracy of the fifteen methods of the main design: geometric mean over the 36 scenarios of each method's ratio to the best method in a scenario, and the largest such ratio, with 95% bootstrap intervals. Circles are foundation models, squares classical methods; lower and further left is better on both counts.

The stronger classical benchmarks do not change the picture. DOTM and the Comb benchmark share Theta's failure on D4, where a decomposition with fixed seasonal indices meets a seasonal pattern that evolves, and replacing Theta by DOTM leaves the combination's worst ratio at 2.99. TimesFM-3 wins 28 of 29 significant comparisons against DOTM and all 29 against the Comb benchmark. **Among the five foundation models, the robustness belongs to TimesFM-3 alone:** TimesFM-2.5 has a worst ratio of 2.18 (on D7 at $n = 24$), and 2.26 with its settings switched off, so the difference is not one of configuration; TiRex has 2.24, Chronos-2 2.75 and Chronos-Bolt 3.06, against 2.22 for AutoARIMA. In the median scenario every foundation model but Chronos-Bolt is within 2% to 4% of the best, and TimesFM-2.5 is the best

method in 7 scenarios, more than any model but AutoARIMA (8). The newer models do not clearly beat AutoARIMA either (Chronos-2 10–10, TiRex 9–12). Friedman tests with Nemenyi critical differences [63] within each regime agree: TimesFM-3 is tied for the best mean rank on the linear processes (with TimesFM-2.5, TiRex and AutoARIMA) and on the break and saturation processes, and AutoARIMA has the best mean rank on the seasonal processes. Averaged over the design, the 80% intervals of TimesFM-3 cover 0.814 (Monte Carlo SE 0.003), TimesFM-2.5 0.812, Chronos-2 0.789, TiRex 0.842, Chronos-Bolt 0.834 and AutoARIMA 0.794.

5.4 | Intermittent demand re-examined

Six purpose-built intermittent-demand methods were added: Croston’s classic and optimised variants, the Syntetos–Boylan approximation [64], ADIDA [65], IMAPA [66] and TSB [67], the last with StatsForecast’s fixed constants $\alpha_d = \alpha_p = 0.2$ (Supporting Information, Table S7). TimesFM-3 beats all six on MASE at every length (24/24 significant; median ratios 0.70 to 0.82), while the Croston family has the lower RMSSE (0.694–0.769, against 0.774–0.806 for TimesFM-3). Because MASE rewards the conditional median, which on D8 is near zero, the trivial all-zero forecast also belongs in the comparison, and because D8 is independent over time, so do benchmarks that use the empirical distribution of the history [68]: its mean as point forecast and its deciles as predictive distribution. Table 5 adds them, with the scaled mean error (bias) and periods in stock [PIS; 69], which accumulates forecast errors as a stock position would.

TABLE 5 | Intermittent demand (D8) re-examined with simple benchmarks, bias and stock measures.

Method	MASE	RMSSE	sME	sPIS	SPL
All-zero forecast	0.668–0.718	0.785–0.825	−0.72 to −0.67	−91 to −81	–
History: mean; empirical deciles	0.924–0.975	0.692–0.744	−0.07 to −0.02	−13 to −3	0.282–0.314
TimesFM-3	0.683–0.780	0.774–0.806	−0.67 to −0.57	−81 to −74	0.288–0.325
TimesFM-3, mean of deciles	0.915–1.002	0.694–0.760	−0.06 to −0.01	−10 to −4	0.288–0.325
TimesFM-2.5	0.678–0.770	0.777–0.806	−0.67 to −0.59	−81 to −74	0.289–0.324
Chronos-Bolt	0.708–0.786	0.758–0.789	−0.59 to −0.54	−72 to −65	0.294–0.328
Chronos-2	0.681–0.750	0.776–0.808	−0.66 to −0.63	−81 to −76	0.290–0.323
TiRex	0.683–0.776	0.771–0.800	−0.67 to −0.58	−81 to −73	0.289–0.324
Croston–SBA	0.921–1.025	0.694–0.763	−0.04 to 0.05	−7 to 2	–
ADIDA	0.926–0.972	0.700–0.753	−0.09 to −0.02	−16 to −5	–
TSB	0.916–0.971	0.716–0.764	−0.10 to 0.00	−17 to −1	–
AutoARIMA	0.924–0.996	0.698–0.765	−0.05 to −0.03	−12 to −3	0.341–0.380

Note: Process D8, 200 replications per length; ranges over the four lengths $n \in \{24, 48, 96, 200\}$. sME: mean error (forecast minus actual) scaled by the MASE denominator; negative values mean under-forecasting. sPIS: periods in stock [69], cumulated over the horizon and scaled by the mean demand of the context; large negative values mean persistent stock-outs. SPL: scaled pinball loss over the nine deciles, defined only for methods with a predictive distribution. TimesFM-3, mean of deciles, uses the average of its nine deciles as point forecast. History: the mean of the context as point forecast and the empirical deciles of the context as predictive distribution; the median of the context is zero or near it, so its point forecast would repeat the all-zero row.

The all-zero forecast has the lowest MASE at every length (0.668 to 0.718), below TimesFM-3’s median forecast (0.683 to 0.780). The MASE advantage of Section 4.4 is thus the advantage of a forecast close to zero: its bias (−0.67 to −0.57 MASE units) and PIS (−81 to −74) are close to those of the all-zero forecast and imply persistent stock-outs. With the mean of its deciles as point forecast, TimesFM-3 has an RMSSE of 0.694 to 0.760 and a bias within 0.06 of zero, level with the best Croston-family method and with the mean of the history (0.692 to 0.744). Its pinball loss, 0.288 to 0.325, is about 15% below that of the classical methods (0.341 to 0.380), but their quantiles come from Gaussian intervals that put probability on negative demand; the empirical deciles of the history score 0.282 to 0.314, slightly lower than TimesFM-3 at every length. **On intermittent demand, therefore, TimesFM-3 shows no advantage over simple benchmarks:** it behaves much like the empirical distribution of the history, and so do the other foundation models (Table 5). A model built for intermittent demand, iETS with automatic occurrence selection [70], has a pinball loss of 0.283 to 0.306, lower than TimesFM-3’s at every length (paired Wilcoxon $p < 0.001$) and level with the empirical deciles: it selects a fixed occurrence probability for 791 of 797 series, which brings its predictive distribution close to that of the history. Negative-binomial and TSB-type predictive distributions fitted to the context do worse than the empirical deciles (Supporting Information, Table S15). The result concerns a process that is independent over time, the case most favourable to history-based benchmarks.

5.5 | Randomised parameters and departures from the assumptions

Every series now draws its own parameters uniformly from wide ranges containing the fixed values of Table 2 (Supporting Information, Table S8): autoregressive coefficients up to 0.95, intermittency from lumpy ($p = 0.1$) to nearly continuous ($p = 0.6$), breaks of three to fifteen noise standard deviations in either direction. Four versions use the same 7 200 parameter draws (200 per process and length): *random parameters* with Gaussian innovations; *heavy tails*, Student- t_3 innovations of the same variance built from the same normal draws (negative binomial sizes on D8); *outliers*, each context point shifted with probability 0.03 by four to six robust standard deviations (a demand multiplied by five on D8), with the held-out values left clean; and *tested seasonality*, in which the classical methods use $m = 12$ only if the 90% autocorrelation test of the M4 benchmarks [3] finds seasonality, applied when at least three cycles are observed, and $m = 1$ otherwise. TimesFM-3 never receives a period, so its forecasts are unchanged in the last version. A fifth version, added later, applies the heavy tails and the outliers together, with the same draws.

TABLE 6 | Headline conclusions under randomised parameters and departures from the assumptions.

	Random	Heavy tails	Outliers	Both	Tested seas.
TimesFM-3 scenario wins (of 36)	14	15	16	19	16
TimesFM-3 median ratio to scenario-best	1.029	1.038	1.004	1.000	1.017
TimesFM-3 worst ratio to scenario-best	1.56	1.54	1.49	1.56	1.55
Smallest worst ratio, classical	2.12	2.32	2.09	2.30	1.58
Significant comparisons, won–lost	93–29	100–28	101–27	104–24	100–27
D4, $n \geq 48$: AutoARIMA better (of 3)	3	3	2	2	3
D8: Croston better on RMSSE (of 4)	4	4	4	4	4
D5, $n = 24$: AutoETS / seasonal naive ($m = 12$)	2.03	2.05	1.91	1.95	2.25

Note: 200 replications per (process, length) scenario, 7 200 series per column; every series draws its own parameters from the ranges in Table S8 of the Supporting Information. Ratios use mean MASE over $h = 1, \dots, 12$. Significant comparisons are paired Wilcoxon tests of TimesFM-3 against each of the five general-purpose classical methods in each scenario (180 per column), Benjamini–Hochberg at 0.05 within each (column, opponent) family of 36 scenarios (the main design’s families also include the three horizon slices). The D8 row compares TimesFM-3 with the best of Croston-SBA, TSB and ADIDA; the last row divides by seasonal naive with the true period in every column. “Both” combines the heavy tails and outliers of the two preceding columns (added later, same draws). In the tested-seasonality column the classical methods use $m = 12$ only where the M4 seasonality test finds seasonality; TimesFM-3’s forecasts are those of the first column.

In the four versions with a known period the conclusions were qualitatively consistent with the main design (Table 6), although TimesFM-3’s worst ratio lies outside the main design’s bootstrap interval: it is the best method in 14 to 19 scenarios, within 4% of the best in the median scenario, and its worst ratio rises to 1.49 to 1.56 (bootstrap intervals up to about 1.83), always on the logistic process D7 at $n = 96$, while the smallest classical worst ratio is 2.09 to 2.32. TimesFM-3 wins 93 to 104 of 180 comparisons and loses 24 to 29, AutoARIMA again being the opponent it does not clearly beat. AutoARIMA’s advantage on D4 with at least four cycles persists (median over the three lengths 21% to 26%; with outliers it is ahead at two of the three lengths, by 15% and 5%, and with outliers and heavy tails together by 16% and 6%), and the Croston family keeps the lower RMSSE on D8 in every version.

When the classical methods test for seasonality, the gap in worst ratios closes, but not because they forecast better: their smallest worst ratio is 1.58 (1.48 to 1.67, AutoARIMA), level with TimesFM-3’s 1.55 (1.41 to 1.71). AutoARIMA’s worst scenario is still D4 at $n = 24$, and its mean MASE there is unchanged (2.30); what changes is the yardstick. At $n = 24$ the test cannot be applied, every series is treated as non-seasonal, and seasonal naive loses the true period that made it the best method in that scenario (mean MASE 4.26 instead of 1.09), so TimesFM-3 (1.45) becomes the scenario-best. Much of the classical deficit in worst ratios is thus measured against seasonal naive with a known period.

A response-surface regression of the per-series $\log_2$ ratio of TimesFM-3’s MASE to AutoARIMA’s on the standardised parameters, $\log_2 n$ and the version, with standard errors clustered by parameter draw (Supporting Information, Tables S4 and S5, and Figure S2), shows that the dependence on the settings is modest. The largest systematic effect is length: each doubling of n moves the ratio against TimesFM-3 by about 18% on D4 and 10% on D5, and in its favour on D6 and D8. Outliers, alone or with heavy tails, favour TimesFM-3 on the seasonal processes (14% and 10%; 14% and 8%). Among the parameters only the demand probability of D8 has a large effect; the ratio is flat in the autoregressive coefficient of D1 and the GARCH persistence of D9, grows in favour of TimesFM-3 with the size of the break in D6, and varies with the inflection point of D7 in a way a linear term does not capture. The regressions explain little of the per-series variation ($R^2 \leq 0.23$).

Familiarity. Because TimesFM-3 is pre-trained partly on synthetic series, its advantage could reflect familiarity with some process families rather than their difficulty. In the randomised design with Gaussian innovations, at matched difficulty (spectral entropy and length), its advantage over AutoARIMA is about 4% to 6% larger on the processes that no classical family specifies (D6–D8) than on D1–D5 (shift in the $\log_2$ ratio -0.096 , 95% interval -0.128 to -0.064 ; median regression -0.066), which is consistent with an effect of familiarity but does not establish one: on D6–D8 AutoARIMA is also misspecified by construction, which predicts the same shift.

5.6 | A longer horizon

Calibration of foundation models is reported to degrade with the horizon [18], and the main design stops at 12 steps. The nine processes were therefore regenerated with the same seeds for a 48-step horizon and forecast by the six pre-registered methods, TimesFM-2.5 and Chronos-2 (Supporting Information, Section S7, Table S14 and Figure S3). Because the inflection of D7 sits at $0.6(n+H)$, the longer horizon moves it into the context for $n = 200$, so that the series has already flattened when the forecast starts. Over the first 12 steps the picture of Section 4 is recovered: without D7, TimesFM-3's worst ratio to the best method is 1.31 and AutoARIMA's 2.11; over steps 13 to 24 TimesFM-3 still has the smaller worst case (1.39 against 1.60). Over steps 25 to 48 it does not: **AutoARIMA has the smaller worst case** (1.41 against 1.80, D7 excluded), although TimesFM-3 remains closer to the best method in the median scenario (1.064 against 1.092, D7 excluded). On D7 at $n = 200$ all three foundation models forecast a decline while the process stays at its plateau near 200: TimesFM-3's mean forecast falls to 128 by step 48, TimesFM-2.5's to 98 and Chronos-2's to 146, and their mean MASE over the 48 steps is 8.37, 12.22 and 5.30, against 1.11 for seasonal naive and 2.51 for AutoARIMA. What in the models produces the decline cannot be established here. On these regenerated series the coverage of TimesFM-3's 80% intervals falls from 0.788 over steps 1 to 12 to 0.760 over steps 25 to 48, and AutoARIMA's from 0.760 to 0.720, so both drift away from nominal, as Adler, Chang, Draxler, Abdi, and Smyth [18] report for foundation models. The robustness of Section 4.3 is thus a property of horizons up to about two years of monthly steps.

6 | Application to real data

Whether the foundation models have seen the M4 data cannot be settled. TimesFM-3's documented real-world corpus, the GIFT-Eval pre-training collection with Wikipedia page views and Google Trends, is built to exclude the GIFT-Eval test datasets, of which M4 Monthly is one [33, 9, 71], but the exclusion works at the level of datasets and the synthetic part is undocumented; TimesFM-2.5 lists the same sources [59]. The original Chronos models were trained on M4 Monthly with its final 18 observations, the official test period, held out [5, Table 2], so Chronos-Bolt's M4 forecasts may be in-domain rather than zero-shot; the corpora of Chronos-2 and TiRex were not checked. The M4 tier is therefore secondary evidence; its official-test and truncation analyses are post hoc, and Section 6.5 adds, post hoc, data observed after the documented training corpora.

6.1 | Representativeness of the simulated series

Following Kang, Hyndman, and Smith-Miles [23], each series is summarised by four features: normalised spectral entropy (forecastability), STL trend strength, STL seasonal strength (period 12) and the first autocorrelation of the first differences. The simulated series at $n = 96$ and 1 000 M4 Monthly series (Section 6.2), each cut to the last 96 observations of its training period, are placed in this space with the features standardised on the M4 series. An M4 series is *covered* when some simulated series lies at least as close to it as the 95th percentile of the distance from an M4 series to its nearest other M4 series.

The fixed design covers 48.5% of the M4 series (bootstrap interval 45.2% to 51.4%; Figure 6): each process forms a compact cluster and the regions between them, which hold many real series, are empty. The randomised design of Section 5.5 covers 75.8% (73.2% to 78.4%). Across thresholds from the 80th to the 99th percentile, lengths of 48, 96 and 200, and a six-feature space adding the first autocorrelation and lumpiness, the randomised design covers 76% to 89% of M4 at the 95th percentile and never less than the fixed design, also when the latter is subsampled to equal size; conversely, 60% to 92% of the randomised series lie within the same distance of some M4 series, fewest at $n = 200$. The parameter ranges were set before this statistic was first computed. M4 series are more strongly trending than the simulated ones (median trend strength 0.87 against 0.63), and only 1.5% lie nearest to D8, because M4 has no intermittent series; the intermittent-demand results rest on the simulation alone. Coverage of the feature space does not mean that results transfer series by series:

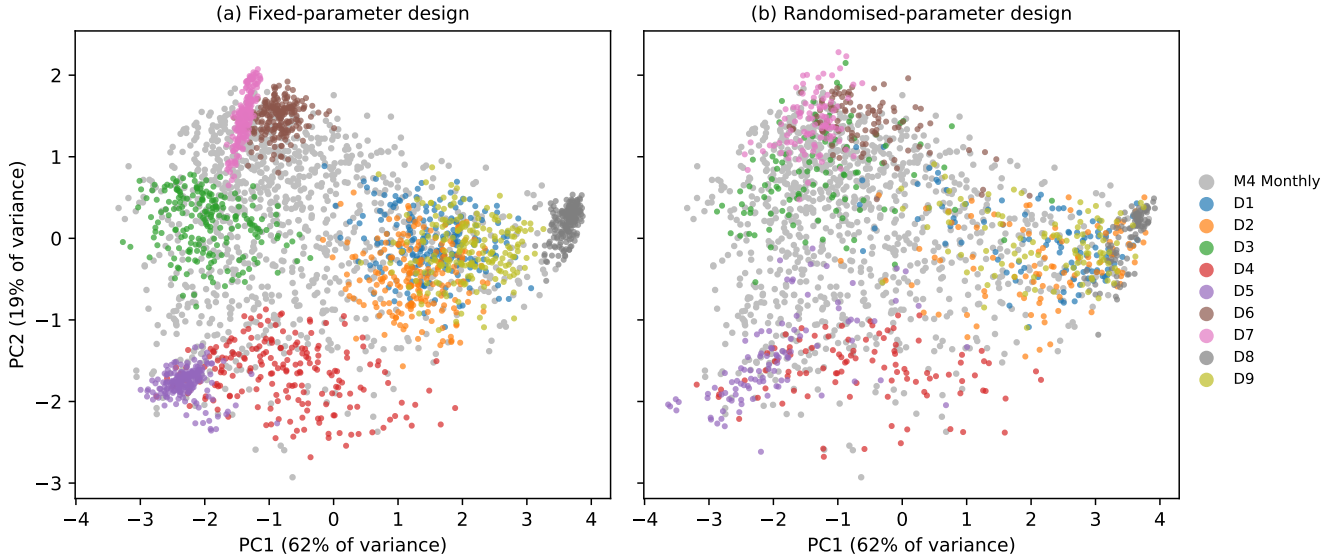


FIGURE 6 | Simulated and M4 Monthly series in a four-feature instance space (spectral entropy, trend strength, seasonal strength and first-order autocorrelation of the differences), first two principal components. Grey points are M4 series; coloured points are simulated series at $n = 96$. (a) The fixed-parameter design covers 48.5% of the M4 series; (b) the randomised-parameter design covers 75.8%, with coverage defined in Section 6.1.

predicting each M4 series' $\log_2$ ratio of TimesFM-3's MASE to AutoARIMA's from its ten nearest simulated series gives a Spearman correlation of 0.06 (95% interval -0.005 to 0.125) with the realised ratio, no better than a constant prediction.

6.2 | Rolling-origin evaluation

A seeded sample of 1 000 series is drawn from the 32 581 M4 Monthly series with at least 120 training observations. Their training periods have a median of 281 observations (interquartile range 198 to 306), against 202 (82 to 306) for all 48 000 series, so the sample over-represents the long end of the population. Each series is evaluated at up to three origins 12 months apart with M4's horizon $h = 18$ (2 932 windows), the latest being the official test period, forecast from the training series; consecutive windows overlap by six months, so the errors of a series are averaged over its origins and the series are the independent units of the paired Wilcoxon tests. TimesFM-3 (mean MASE 0.914) significantly outperforms seasonal naive (1.280; better on 80.1% of series), Theta (0.970; 56.8%) and AutoETS (0.951; 54.1%), and shows no detectable difference from AutoARIMA (0.923; 52.6%, adjusted $p = 0.27$) or the combination (0.907; 48.5%, adjusted $p = 0.27$). This ties TimesFM-3 with AutoARIMA on long, mostly seasonal series, rather than showing the AutoARIMA advantage found on the simulated D4: real series mix trend, seasonality and irregular components in proportions the simulation does not reproduce (Section 6.1), and an advantage from contamination cannot be excluded either. The probabilistic results also diverge: TimesFM-3's 80% intervals cover 0.765 against 0.801 for the combination, yet its pinball loss is the lowest of the six methods (0.363, against 0.365 for the combination and 0.374 for AutoARIMA)—a better distribution overall with too narrow an 80% interval, the fragility of foundation-model calibration reported by Adler, Chang, Draxler, Abdi, and Smyth [18].

6.3 | The official M4 test period

To place the results against the published M4 results, the official 18-month test period, the latest of the three origins above, was scored as in M4, with the overall weighted average (OWA) of sMAPE and MASE relative to the official Naive2 forecasts [3]. The comparison includes the methods of Section 5.3 and the published point forecasts of the ten best M4 submissions, among them the winning hybrid of exponential smoothing and a recurrent network [72] and the feature-based combination FFORMA [73]; scored on all 48 000 monthly series, the downloaded files reproduce the published M4 results.

Theta and the Comb benchmark improve on Naive2 by 7% to 9% in OWA (Table 7), in line with the competition. **Four M4 entries are ahead of every foundation model:** FFORMA (OWA 0.831), Jaganathan and Prakash (0.841), Smyl's hybrid (0.848) and Pawlikowski et al. (0.849). TimesFM-3 (0.862; 0.857 with its evaluator settings) is level with the entries ranked

TABLE 7 | M4 Monthly official test period: accuracy, OWA and multiple comparisons.

Method	sMAPE	MASE	OWA	Coverage	MSIS	Mean rank
FFORMA (M4, 2nd)	7.82	0.889	0.831	–	–	9.85*
Jaganathan & Prakash (M4, 4th)	7.88	0.902	0.841	–	–	10.06*
ES-RNN, Smyl (M4, 1st)	7.98	0.907	0.848	–	–	10.72*
Pawlikowski et al. (M4, 3rd)	7.95	0.912	0.849	–	–	10.38*
TiRex	8.04	0.916	0.855	0.801	4.40	10.77*
TimesFM-3, evaluator settings	8.01	0.923	0.857	0.765	4.50	–
Petropoulos & Svetunkov (M4, 6th)	8.09	0.916	0.858	–	–	10.46*
Chronos-2	8.11	0.917	0.860	0.783	4.47	10.69*
Fiorucci & Louzada (M4, 5th)	8.09	0.920	0.860	–	–	10.50*
TimesFM-3	8.05	0.929	0.862	0.761	4.54	10.69*
TimesFM-2.5	8.15	0.922	0.864	0.812	4.53	11.03
Shaub (M4, 7th)	8.06	0.933	0.864	–	–	11.01
Combination (ETS, ARIMA, DOTM)	8.16	0.928	0.867	0.789	4.74	–
Combination (Theta, ETS, ARIMA)	8.16	0.929	0.868	0.789	4.76	10.58*
TimesFM-2.5, settings off	8.25	0.932	0.874	0.802	4.61	–
Doornik et al. (M4, 9th)	8.21	0.945	0.878	–	–	11.64
Pedregal et al. (M4, 10th)	8.31	0.953	0.887	–	–	11.88
AutoARIMA	8.50	0.939	0.891	0.779	4.91	11.13
Chronos-Bolt	8.40	0.971	0.900	0.800	4.70	11.91
AutoETS	8.45	0.966	0.900	0.784	4.98	11.82
M4 Comb	8.50	0.982	0.911	–	–	12.65
Legaki & Koutsouri (M4, 8th)	8.51	0.995	0.917	–	–	12.76
DOTM	8.59	0.996	0.922	0.728	5.27	13.03
Theta	8.59	1.006	0.926	0.723	5.36	13.00
Naive2	9.22	1.092	1.000	–	–	–
Seasonal naive	10.39	1.306	1.161	0.780	6.56	16.44

Note: 1 000 M4 Monthly series, each forecast for the official 18-month test period from its full training history. sMAPE and MASE as defined in M4; OWA relative to the official Naive2 forecasts. Coverage and MSIS use the nominal 80% interval, the widest that every probabilistic method can form from nine deciles. Mean rank: average rank of per-series MASE among 22 methods, one configuration per model (Friedman $\chi^2 = 1054$, $p < 10^{-100}$); * marks methods whose mean rank is within the Nemenyi critical difference (1.04) of the best; – marks alternative configurations and Naive2, left out of the ranking. Published M4 submissions, ranked by their overall M4 position, provide point forecasts only [3, 72, 73].

fifth to seventh in M4 (0.858 to 0.864), TiRex (0.855) and Chronos-2 (0.860) are close to it, and the classical combination (0.868) and AutoARIMA (0.891) follow. On this sample the M4 entries' OWA differs from their published monthly values by up to 0.03, as much as the gaps between them. A Friedman test on per-series ranks, with one configuration per model and without Naive2 [74], rejects the equality of the 22 methods, but ten lie within the Nemenyi critical difference of the best mean rank [63]: six M4 entries, the classical combination, TimesFM-3, Chronos-2 and TiRex. On real monthly series with long histories, zero-shot TimesFM-3 matches good specialised methods of 2018 without per-series tuning, but not the best of them. Reweighting the sample to the length distribution of the population it was drawn from leaves TimesFM-3's OWA at 0.861 and the four leading entries unchanged; the shorter third of the M4 Monthly series is not represented, and the truncation experiment below stands in for it.

6.4 | Truncated histories

The official test period was also forecast from only the last 24, 48 or 96 observations of each training series, with the full-history MASE denominator so that only the information available to the methods changes (Table 8). The simulation's estimability result reappears: with 24 observations, where the classical methods again have two cycles for a period of 12, the foundation models lead—TiRex 1.027, TimesFM-3 1.046, TimesFM-2.5 1.050—against 1.096 for the best classical method, the combination, and 1.175 for AutoARIMA; the difference was not tested. With 48 observations the combination is level with them (0.955 against 0.952 to 0.972), and with the full history the leading methods are within 2% of one another. Truncation also moves the information set away from the full series a model could have been trained on, although it cannot rule out that the series were seen.

TABLE 8 | M4 Monthly with truncated contexts: mean MASE by context length.

Method	Last 24	Last 48	Last 96	Full
TimesFM-3	1.046	0.961	0.942	0.929
TimesFM-3, evaluator settings	1.045	0.960	0.936	0.923
TimesFM-2.5	1.050	0.972	0.952	0.922
Chronos-Bolt	1.158	1.023	0.965	0.971
Chronos-2	1.087	0.952	0.935	0.917
TiRex	1.027	0.956	0.949	0.916
AutoARIMA	1.175	1.019	0.979	0.939
AutoETS	1.129	0.996	0.995	0.966
Theta	1.150	1.011	1.003	1.006
DOTM	1.167	1.012	0.991	0.996
Combination (Theta, ETS, ARIMA)	1.096	0.955	0.947	0.929
Combination (ETS, ARIMA, DOTM)	1.111	0.959	0.945	0.928
Seasonal naive	1.306	1.306	1.306	1.306

Note: Mean MASE on the M4 official test period when each method sees only the last 24, 48 or 96 training observations, or the full history (median 281 observations). All columns use the full-history MASE denominator, so only the information given to the methods changes. Seasonal naive uses only the last 12 observations and is unaffected.

6.5 | Data observed after the training corpora

Neither the simulation nor M4 can show how the foundation models behave on real data they cannot have memorised. The monthly FRED-MD database [75], in its August 2026 vintage, offers such data: 101 of its 126 series are complete from 1990 to June 2026, and every method forecast January 2025 to June 2026 (18 steps) from the 240 months to December 2024, raw levels without transformation. The evaluation window post-dates the documented corpora of TimesFM-3 and TimesFM-2.5 (to November 2023) and the releases of Chronos-Bolt; TiRex and Chronos-2 were released within it, and restricting the window to November 2025 onwards leaves AutoARIMA first (mean MASE 0.722) and puts TimesFM-3 behind every method except the naive benchmarks (0.857). On this tier **TimesFM-3 is statistically level with the classical methods** (Supporting Information, Table S16): AutoARIMA has the lowest mean MASE (0.540), TiRex and the combination follow (0.556), and TimesFM-3 has 0.607, with a median (0.426) close to AutoARIMA's (0.411); paired Wilcoxon tests with Benjamini–Hochberg correction separate TimesFM-3 only from seasonal naive, which it beats, and a Friedman test puts every method except seasonal naive within the Nemenyi critical difference of the best mean rank. TimesFM-3's mean is raised by a few series on which it extended a recent movement far beyond what followed, such as the 2024 cuts in short-term interest rates. The tier is small and dominated by trending macroeconomic levels, and it rules out memorised test values, not familiarity with the series' past.

7 | Discussion

7.1 | Summary of findings

Table 9 collects the findings with their evidence, their status with respect to the protocol and the conditions under which they hold.

7.2 | Robustness versus peak accuracy

The classical methods' 20 scenario wins are spread over five methods, so which classical method wins is not knowable in advance without the model selection a foundation model is meant to remove. TimesFM-3, one untuned model, is not significantly worse than the best classical method in 25 of the 36 scenarios; the best classical method is ahead in the other 11, over five processes, by small margins on D2 and D3 and by large ones only on D4, D5 and D7 at $n = 96$ (these comparisons with a selected opponent are descriptive). The evidence therefore supports neither the conclusion that foundation models have superseded classical methods nor the opposite, but a narrower one: **in this design TimesFM-3 behaved robustly rather than with peak accuracy**. It is rarely the best method and never among the worst, whereas the automatic classical methods are sometimes excellent and sometimes catastrophic. For one well-understood series that

TABLE 9 | Summary of findings.

Finding	Evidence	Status	Holds within
Neither family dominates; AutoARIMA is the only classical method TimesFM-3 does not clearly beat	Section 4.1, Figure 3	Pre-registered	Nine processes, $H = 12$
TimesFM-3's worst ratio to the best method is 1.31, every classical method's at least 2.2	Table 3, Section 5.1	Post-hoc summary	$H = 12$; relative to the methods compared
The largest failures of automatic ARIMA and ETS come from seasonal models that two cycles cannot estimate	Section 4.3, Section 5.2	Pre-registered, checked post hoc	$m = 12, n = 24$; reproduced in R
With four or more cycles AutoARIMA is 21–24% more accurate on seasonal ARIMA data	Section 4.4	Pre-registered	D4, D5
The robustness is specific to TimesFM-3 among five foundation models	Section 5.3, Figure 5	Post hoc	$H = 12$
On intermittent demand no advantage over simple history-based benchmarks	Section 5.4	Post hoc	An independent process
At steps 25–48 AutoARIMA has the smaller worst case; the three foundation models run there forecast a decline on a plateau	Section 5.6	Post hoc	Regenerated design
On M4, TimesFM-3 is level with the fifth- to seventh-ranked entries, behind the four best	Section 6.3	Post hoc	1 000 long monthly series
On data observed after the documented training corpora, TimesFM-3 is level with the classical methods	Section 6.5	Post hoc	101 FRED-MD series, 2025–2026

trade is unattractive; for many heterogeneous series and no capacity to diagnose each one, it favoured TimesFM-3 here, within the horizons, processes and yardstick set out in Table 9.

7.3 | Practical considerations

The considerations below are conditional on the processes, lengths and horizons examined. They describe how the methods behaved here and are offered as diagnostic starting points to be checked on the user's own data, not as rules.

Many heterogeneous series without per-series tuning. This is where TimesFM-3's robustness mattered most: it was never far from the best method of a scenario. Where a model can be identified for a single series, TimesFM-3 gains little over AutoARIMA, which keeps parameters, diagnostics and interval theory that can be explained and audited; when TimesFM-3 forecasts poorly, there is nothing to inspect, and its quantiles are learned outputs with no accompanying theory, well calibrated on average in the simulation but not on M4. In regulated settings where a forecast must be justified, these differences may outweigh the accuracy comparison.

Seasonal series. With at least four observed cycles AutoARIMA was clearly ahead on seasonal ARIMA data, and seasonal naive was also ahead of TimesFM-3 there. With two cycles the automatic classical methods fell back to non-seasonal models and failed, and seasonal naive, or TimesFM-3, was the safer choice.

Intermittent demand. Simple benchmarks built from the history were as good as anything tested: the empirical deciles of the history gave the lowest pinball loss, and its mean, like the Croston family, the lowest RMSSE, while TimesFM-3's median forecast was no better than forecasting zero.

Long horizons. Beyond about two years of monthly steps AutoARIMA had the smaller worst case, and the three foundation models run there forecast a decline on a process that had levelled off, so long-horizon forecasts from these models should be checked against the recent level before use.

Cost. Timed on the same 200 seasonal series of length 96, after a warm-up run, seasonal naive took 2.8 ms per series, Theta 9.1 ms, AutoETS 106.7 ms and AutoARIMA 2 369.7 ms, each on one CPU core, and TimesFM-3 24.9 ms batched on one consumer GPU and 47.1 ms on the CPU (12 threads), excluding a one-off model load of 2 to 4 s, so TimesFM-3 does not require a GPU. Without the warm-up, which leaves in the one-off compilation of the StatsForecast routines, the classical timings differ by less than 10%. In the main run the four classical methods took 7.3 minutes on 22 cores against 3.2 minutes for TimesFM-3, a ratio of 2.3× that is dominated by AutoARIMA and depends on the hardware. The classical methods hold a handful of parameters per series; TimesFM-3 keeps a checkpoint of about 1.3 GB resident, whatever the number of series.

Licence. TimesFM-3's weights are released under the TimesFM Non-Commercial License v1.0, which grants use “solely for your Non-Commercial Purposes” and requires a separate licence for any “commercial or production activity” [34, 33] (accessed 23 September 2026). The source code is Apache-2.0, as are the weights of TimesFM-2.5, Chronos-Bolt and

Chronos-2, but the version-3 weights are not: **for a commercial application, TimesFM-3 is not licensed as released**, whatever its accuracy, and the alternatives are TimesFM-2.5 (less robust in the simulation, level on M4), separate terms or a classical method. Licensing is routinely omitted from model comparisons although it decides which models an entire class of users may use.

7.4 | Threats to validity

Construct. The summaries that carry the robustness finding—worst ratios and win counts—were defined after the protocol and depend on the methods and scenarios compared, notably on seasonal naive with the known period; on D8 the verdict depends on the error measure. *Internal.* No method was tuned by hand, and the classical methods were run in StatsForecast at its defaults; the cross-check with forecast reproduced the failures but found one implementation failure, and an expert who identified each process could select better models. The protocol was frozen before any forecast but held in the replication archive rather than deposited with an external registry, so its only external timestamp post-dates the forecasts. Everything in Section 5 and Section 6 except the rolling-origin evaluation is post hoc. Only the data of Section 6.5 are known to post-date the documented training corpora. *External.* Simulated series are clean and free of the measurement error, calendar effects and regime changes of applied work; the M4 tier addresses this only in part, and the randomised design still leaves about a quarter of M4 monthly series, and all other frequencies, outside the regions studied. The nine process families are fixed, the seasonal period is 12, the pre-registered horizon is 12 steps and the longer one was examined in one secondary check, and the departures from the assumptions were applied one at a time and in one combination. TimesFM-3 was used zero-shot and univariate, leaving its multivariate and covariate capabilities untested; fine-tuning on one series of 24 to 200 points is not meaningful for a model of this size, and fine-tuning on the DGPs would remove the separation between model and test data. Five foundation models were evaluated and they behave differently, so conclusions about one do not transfer to the class; other families, such as Moirai [6, 16], Sundial, Time-MoE and Toto, were not included.

8 | Conclusion

Under a pre-registered protocol and on data no model could have seen, TimesFM-3 and classical methods were compared on 7 200 series from nine known processes. Neither family dominates: TimesFM-3 wins most significant comparisons, but not against AutoARIMA, and with at least four seasonal cycles AutoARIMA is clearly more accurate. What distinguishes TimesFM-3 is its worst case, an error never more than 1.31 times that of the best method in a scenario against at least 2.2 times for every classical method, and among five foundation models it is the only one with this property. The property is bounded: it is measured relative to the methods compared, it holds for horizons up to about two years of monthly steps, and on intermittent demand simple benchmarks built from the history do as well. On the M4 test period TimesFM-3 is level with good but not the best M4 entries, and on macroeconomic data observed after its documented training data it is level with the classical methods. The finding most likely to transfer concerns the classical side, where the mechanism can be examined: what matters is whether a classical model is *estimable* from the data, not only whether it is correctly *specified*.

The study is exploratory. It describes how one black-box model behaves on data of known structure, does not explain that behaviour, and remains conditional on the processes, parameter ranges, horizons and benchmarks examined. For commercial users the non-commercial licence of TimesFM-3's weights decides the question before accuracy does.

9 | Computational details and reproducibility

All results were produced with Python 3.13 [76], statsforecast 2.1.1 [45], timesfm 3.0.1 with the google/timesfm-3.0-pytorch weights (revision 43046b8) [34], chronos-forecasting 2.3.2 with amazon/chronos-bolt-base (revision 5d9f166) and amazon/chronos-2 (revision 29ec376), tirex-ts 1.4.2 with NX-AI/TiRex (revision 63c7409), the google/timesfm-2.5-200m-pytorch weights (revision 1d95242), R 4.4.1 with forecast 9.0.2 and smooth 4.5.2 for the cross-check and iETS, PyTorch 2.6.0 with CUDA 12.4 and cuDNN 9.1 [77], NumPy [78], pandas [79], SciPy [80] and Matplotlib [81], on an NVIDIA GTX 1660 Ti (driver 591.86) with a 24-core CPU and 32 GB of memory; parallel classical runs use 22 worker processes. The full package list is pinned in the replication archive. Every simulated series is seeded from its own identifiers, so the study regenerates exactly, independently of core count or execution order, and each cached stage can be rerun alone. The foundation models were run in 32-bit arithmetic without random sampling; rerunning two TimesFM-3 scenarios reproduced the stored quantiles exactly, although other hardware may differ in the last digits because GPU

floating-point reductions are not associative. The protocol, with its five hypotheses, was last modified on 8 September 2026 at 20:01, 22 minutes before the first forecast file was written (local file-system timestamps); its first external timestamp is the public archive of 9 September 2026, which postdates the forecasts, and it has not been edited since. The archive holds a dated log of the fifteen departures from it, each reasoned, and every post-hoc analysis is marked as such. Every DOI in the bibliography was resolved and compared with its entry.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest.

ETHICS APPROVAL

Not applicable; the study uses simulated data and publicly available data (M4 competition data and forecasts, FRED-MD), and involves no human participants. Permission to reproduce material from other sources: not applicable.

DATA AVAILABILITY STATEMENT

The complete research compendium is archived on Zenodo, <https://doi.org/10.5281/zenodo.22681072> (concept DOI, resolving to all versions), and mirrored at <https://github.com/ebrahimkhaled/timesfm-classical-paper>: the pre-registered protocol, the deviation log, the entire pipeline, the per-series metrics, the raw forecasts and quantiles, every figure and the manuscript source. The simulation data are generated by that code from documented seeds; the real-data tier uses the publicly available M4 competition data [3], which the code re-fetches, with the seeded 1 000-series sample included, and the FRED-MD database (August 2026 vintage) [75] and the published M4 forecasts, which are publicly available and are fetched by the code rather than archived. The raw forecasts are deposited so that they can be re-scored under other losses. TimesFM-3's weights are not redistributed; the code downloads them from Hugging Face.

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References

1. Ayush Jain and Rajat Sen. 2026. "TimesFM-3: A Zero-Shot Foundation Model for Multivariate Forecasting." Google Research Blog. Released 31 August 2026.
2. Marcel Meyer, Sascha Kaltenpoth, Kevin Zalipski, and Oliver Müller. 2025. "Rethinking Evaluation in the Era of Time Series Foundation Models: (Un)known Information Leakage Challenges." Preprint, arXiv:2510.13654; no published version as of September 2026.
3. Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. "The M4 Competition: 100,000 Time Series and 61 Forecasting Methods." *International Journal of Forecasting* 36, no. 1 (2020): 54–74.

4. Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. 2024. “A Decoder-Only Foundation Model for Time-Series Forecasting.” In *Proceedings of the 41st International Conference on Machine Learning*, Vol 235 of *Proceedings of Machine Learning Research*, 10148–10167. PMLR. Preprint: arXiv:2310.10688.
5. Abdul Fatir Ansari, Lorenzo Stella, Ali Caner Türkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. “Chronos: Learning the Language of Time Series.”. *Transactions on Machine Learning Research*. Preprint: arXiv:2403.07815.
6. Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. 2024. “Unified Training of Universal Time Series Forecasting Transformers.” In *Proceedings of the 41st International Conference on Machine Learning*, Vol 235 of *Proceedings of Machine Learning Research*, 53140–53164. PMLR. Preprint: arXiv:2402.02592.
7. Azul Garza, Cristian Challu, and Max Mergenthaler-Canseco. 2023. “TimeGPT-1.” arXiv preprint arXiv:2310.03589.
8. Kashif Rasul, Arjun Ashok, Andrew Robert Williams, Hena Ghonia, Rishika Bhagwatkar, Arian Khorasani, et al. 2023. “Lag-Llama: Towards Foundation Models for Probabilistic Time Series Forecasting.” arXiv preprint arXiv:2310.08278.
9. Taha Aksu, Gerald Woo, Juncheng Liu, Xu Liu, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. 2024. “GIFT-Eval: A Benchmark for General Time Series Forecasting Model Evaluation.” In *NeurIPS Workshop on Time Series in the Age of Large Models*, Preprint: arXiv:2410.10393.
10. Oleksandr Shchur, Abdul Fatir Ansari, Caner Turkmen, Lorenzo Stella, Nick Erickson, Pablo Guerron, Michael Bohlke-Schneider, and Yuyang Wang. 2025. “fev-bench: A Realistic Benchmark for Time Series Forecasting.” arXiv preprint arXiv:2509.26468.
11. Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, et al. 2024. “MOMENT: A Family of Open Time-Series Foundation Models.” arXiv preprint arXiv:2402.03885.
12. Vijay Ekambaram, Arindam Jati, Pankaj Dayama, Sumanta Mukherjee, et al. 2024. “Tiny Time Mixers (TTMs): Fast Pre-trained Models for Enhanced Zero/Few-Shot Forecasting of Multivariate Time Series.” arXiv preprint arXiv:2401.03955.
13. Xiaoming Shi, Shiyu Wang, Yuqi Nie, Dianqi Li, et al. 2024. “Time-MoE: Billion-Scale Time Series Foundation Models with Mixture of Experts.” arXiv preprint arXiv:2409.16040.
14. Yong Liu, Guo Qin, Zhiyuan Shi, Zhi Chen, et al. 2025. “Sundial: A Family of Highly Capable Time Series Foundation Models.” arXiv preprint arXiv:2502.00816.
15. Ben Cohen, Emaad Khwaja, Youssef Doubli, Salahidine Lemaachi, et al. 2025. “This Time is Different: An Observability Perspective on Time Series Foundation Models.” arXiv preprint arXiv:2505.14766.
16. Chenghao Liu, Taha Aksu, Juncheng Liu, Xu Liu, Hanshu Yan, et al. 2025. “Moirai 2.0: When Less Is More for Time Series Forecasting.” arXiv preprint arXiv:2511.11698.
17. Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. “Statistical and Machine Learning Forecasting Methods: Concerns and Ways Forward.” *PLOS ONE* 13, no. 3 (2018): e0194889.
18. Coen Adler, Yuxin Chang, Felix Draxler, Samar Abdi, and Padhraic Smyth. 2026. “Beyond Accuracy: Are Time Series Foundation Models Well-Calibrated?.” In *Proceedings of the International Conference on Learning Representations*, Preprint: arXiv:2510.16060.
19. Rachel Yui Ki Chu, Kam Chiu Szeto, and Pui Hong Chung. “Multivariate Influenza Forecasting Using Time-Series Foundation Models: A Comparative Study with Classical Statistical Models Using 11 Years of Surveillance Data in Hong Kong.”. *Scientific Reports*. Published online 31 July 2026; volume and article number not yet assigned.
20. Theo Berger. “Deep Learning and Econometric Time Series Analysis: An Assessment of Daily Return Forecasts.” *Journal of Forecasting* 45, no. 1 (2026): 377–390.
21. Betty X. Y. Chu and Cathy W. S. Chen. “Robust Prediction Intervals for Time Series Forecasting: A Bootstrap and Bayesian Approach.” *Journal of Forecasting* 45, no. 5 (2026): 2309–2331.
22. Francesco Lisi and Matteo Grigoletto. “Multiple Seasonal Autoregressive Integrated Moving Average Models.” *Journal of Forecasting* 44, no. 6 (2025): 2037–2052.
23. Yanfei Kang, Rob J. Hyndman, and Kate Smith-Miles. “Visualising Forecasting Algorithm Performance Using Time Series Instance Spaces.” *International Journal of Forecasting* 33, no. 2 (2017): 345–358.
24. Thiyanga S. Talagala, Rob J. Hyndman, and George Athanasopoulos. “Meta-Learning How to Forecast Time Series.” *Journal of Forecasting* 42, no. 6 (2023): 1476–1501.
25. Rob J. Hyndman and Yeasmin Khandakar. “Automatic Time Series Forecasting: The forecast Package for R.” *Journal of Statistical Software* 27, no. 3 (2008): 1–22.
26. Rob J. Hyndman, Anne B. Koehler, Ralph D. Snyder, and Simone Grose. “A State Space Framework for Automatic Forecasting Using Exponential Smoothing Methods.” *International Journal of Forecasting* 18, no. 3 (2002): 439–454.
27. V. Assimakopoulos and K. Nikolopoulos. “The Theta Model: A Decomposition Approach to Forecasting.” *International Journal of Forecasting* 16, no. 4 (2000): 521–530.
28. Spyros Makridakis and Michèle Hibon. “The M3-Competition: Results, Conclusions and Implications.” *International Journal of Forecasting* 16, no. 4 (2000): 451–476.
29. Rob J. Hyndman and Anne B. Koehler. “Another Look at Measures of Forecast Accuracy.” *International Journal of Forecasting* 22, no. 4 (2006): 679–688.
30. Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. “The M5 Competition: Background, Organization, and Implementation.” *International Journal of Forecasting* 38, no. 4 (2022): 1325–1336.
31. Spyros Makridakis, Evangelos Spiliotis, Vassilios Assimakopoulos, Zhi Chen, Anil Gaba, Ilia Tsetlin, and Robert L. Winkler. “The M5 Uncertainty Competition: Results, Findings and Conclusions.” *International Journal of Forecasting* 38, no. 4 (2022):

1365–1385.

32. Hansika Hewamalage, Klaus Ackermann, and Christoph Bergmeir. “Forecast Evaluation for Data Scientists: Common Pitfalls and Best Practices.” *Data Mining and Knowledge Discovery* 37, no. 2 (2023): 788–832.
33. Google Research 2026. “google/timesfm-3.0-pytorch.” Model card, Hugging Face Hub.
34. Google Research 2026. “TimesFM: A Pretrained Time-Series Foundation Model.” Software repository, version 3.0.1. Model weights google/timesfm-3.0-pytorch released under timesfm-non-commercial-license-v1.0; source code Apache-2.0.
35. Tim P. Morris, Ian R. White, and Michael J. Crowther. “Using Simulation Studies to Evaluate Statistical Methods.” *Statistics in Medicine* 38, no. 11 (2019): 2074–2102.
36. Peter J. Brockwell and Richard A. Davis. 2016. *Introduction to Time Series and Forecasting*, Springer Texts in Statistics. Cham: Springer-Verlag.
37. Rob J. Hyndman, Anne B. Koehler, J. Keith Ord, and Ralph D. Snyder. 2008. *Forecasting with Exponential Smoothing*, Springer Series in Statistics. Berlin: Springer-Verlag.
38. M. Hashem Pesaran and Allan Timmermann. “Selection of Estimation Window in the Presence of Breaks.” *Journal of Econometrics* 137, no. 1 (2007): 134–161.
39. Nigel Meade and Towhidul Islam. “Modelling and Forecasting the Diffusion of Innovation – A 25-Year Review.” *International Journal of Forecasting* 22, no. 3 (2006): 519–545.
40. A. A. Syntetos, J. E. Boylan, and J. D. Croston. “On the Categorization of Demand Patterns.” *Journal of the Operational Research Society* 56, no. 5 (2005): 495–503.
41. J. D. Croston. “Forecasting and Stock Control for Intermittent Demands.” *Operational Research Quarterly* 23, no. 3 (1972): 289–303.
42. Tim Bollerslev. “Generalized Autoregressive Conditional Heteroskedasticity.” *Journal of Econometrics* 31, no. 3 (1986): 307–327.
43. Robert F. Engle and Tim Bollerslev. “Modelling the Persistence of Conditional Variances.” *Econometric Reviews* 5, no. 1 (1986): 1–50.
44. Peter R. Hansen and Asger Lunde. “A Forecast Comparison of Volatility Models: Does Anything Beat a GARCH(1,1)?” *Journal of Applied Econometrics* 20, no. 7 (2005): 873–889.
45. Federico Garza, Max Mergenthaler Canseco, Cristian Challú, and Kin G. Olivares. 2022. “StatsForecast: Lightning Fast Forecasting with Statistical and Econometric Models.” Python package, version 2.1.1.
46. Rob J. Hyndman and George Athanasopoulos. 2021. *Forecasting: Principles and Practice*, Melbourne: OTexts.
47. Fotios Petropoulos, Daniele Apiletti, Vassilios Assimakopoulos, et al. “Forecasting: Theory and Practice.” *International Journal of Forecasting* 38, no. 3 (2022): 705–871.
48. Stephan Kolassa. “Evaluating Predictive Count Data Distributions in Retail Sales Forecasting.” *International Journal of Forecasting* 32, no. 3 (2016): 788–803.
49. Tilmann Gneiting and Adrian E. Raftery. “Strictly Proper Scoring Rules, Prediction, and Estimation.” *Journal of the American Statistical Association* 102, no. 477 (2007): 359–378.
50. Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. “Probabilistic Forecasts, Calibration and Sharpness.” *Journal of the Royal Statistical Society B* 69, no. 2 (2007): 243–268.
51. Frank Wilcoxon. “Individual Comparisons by Ranking Methods.” *Biometrics Bulletin* 1, no. 6 (1945): 80–83.
52. J. L. Hodges and E. L. Lehmann. “Estimates of Location Based on Rank Tests.” *The Annals of Mathematical Statistics* 34, no. 2 (1963): 598–611.
53. Yoav Benjamini and Yoel Hochberg. “Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing.” *Journal of the Royal Statistical Society B* 57, no. 1 (1995): 289–300.
54. Francis X. Diebold and Roberto S. Mariano. “Comparing Predictive Accuracy.” *Journal of Business & Economic Statistics* 13, no. 3 (1995): 253–263.
55. David Harvey, Stephen Leybourne, and Paul Newbold. “Testing the Equality of Prediction Mean Squared Errors.” *International Journal of Forecasting* 13, no. 2 (1997): 281–291.
56. Tilmann Gneiting. “Making and Evaluating Point Forecasts.” *Journal of the American Statistical Association* 106, no. 494 (2011): 746–762.
57. Fotios Petropoulos and Ivan Svetunkov. “A Simple Combination of Univariate Models.” *International Journal of Forecasting* 36, no. 1 (2020): 110–115.
58. Jose A. Fiorucci, Tiago R. Pellegrini, Francisco Louzada, Fotios Petropoulos, and Anne B. Koehler. “Models for Optimising the Theta Method and Their Relationship to State Space Models.” *International Journal of Forecasting* 32, no. 4 (2016): 1151–1161.
59. Google Research 2025. “google/timesfm-2.5-200m-pytorch.” Model card, Hugging Face Hub, accessed 23 September 2026.
60. Amazon Science 2024. “amazon/chronos-bolt-base.” Model card, Hugging Face Hub, accessed 23 September 2026.
61. Abdul Fatir Ansari, Oleksandr Shchur, Jari Küken, Andreas Auer, Boran Han, et al. 2025. “Chronos-2: From Univariate to Universal Forecasting.” arXiv preprint arXiv:2510.15821.
62. Andreas Auer, Patrick Podest, Daniel Klotz, Sebastian Böck, Günter Klambauer, and Sepp Hochreiter. 2025. “TiRex: Zero-Shot Forecasting Across Long and Short Horizons with Enhanced In-Context Learning.” arXiv preprint arXiv:2505.23719.
63. Janez Demšar. “Statistical Comparisons of Classifiers over Multiple Data Sets.” *Journal of Machine Learning Research* 7 (2006): 1–30.

64. Aris A. Syntetos and John E. Boylan. “The Accuracy of Intermittent Demand Estimates.” *International Journal of Forecasting* 21, no. 2 (2005): 303–314.
65. Konstantinos Nikolopoulos, Aris A. Syntetos, John E. Boylan, Fotios Petropoulos, and Vassilios Assimakopoulos. “An Aggregate–Disaggregate Intermittent Demand Approach (ADIDA) to Forecasting: An Empirical Proposition and Analysis.” *Journal of the Operational Research Society* 62, no. 3 (2011): 544–554.
66. Fotios Petropoulos and Nikolaos Kourentzes. “Forecast Combinations for Intermittent Demand.” *Journal of the Operational Research Society* 66, no. 6 (2015): 914–924.
67. Ruud H. Teunter, Aris A. Syntetos, and M. Zied Babai. “Intermittent Demand: Linking Forecasting to Inventory Obsolescence.” *European Journal of Operational Research* 214, no. 3 (2011): 606–615.
68. Thomas R. Willemain, Charles N. Smart, and Henry F. Schwarz. “A New Approach to Forecasting Intermittent Demand for Service Parts Inventories.” *International Journal of Forecasting* 20, no. 3 (2004): 375–387.
69. Peter Wallström and Anders Segerstedt. “Evaluation of Forecasting Error Measurements and Techniques for Intermittent Demand.” *International Journal of Production Economics* 128, no. 2 (2010): 625–636.
70. Ivan Svetunkov and John E. Boylan. “iETS: State Space Model for Intermittent Demand Forecasting.” *International Journal of Production Economics* 265 (2023): 109013.
71. Salesforce AI Research 2024. “GiftEvalPretrain.” Dataset card, Hugging Face Hub, accessed 23 September 2026.
72. Slawek Smyl. “A Hybrid Method of Exponential Smoothing and Recurrent Neural Networks for Time Series Forecasting.” *International Journal of Forecasting* 36, no. 1 (2020): 75–85.
73. Pablo Montero-Manso, George Athanasopoulos, Rob J. Hyndman, and Thiyanga S. Talagala. “FFORMA: Feature-Based Forecast Model Averaging.” *International Journal of Forecasting* 36, no. 1 (2020): 86–92.
74. Alessio Benavoli, Giorgio Corani, and Francesca Mangili. “Should We Really Use Post-Hoc Tests Based on Mean-Ranks?.” *Journal of Machine Learning Research* 17, no. 5 (2016): 1–10.
75. Michael W. McCracken and Serena Ng. “FRED-MD: A Monthly Database for Macroeconomic Research.” *Journal of Business & Economic Statistics* 34, no. 4 (2016): 574–589.
76. Python Core Team 2025. *Python: A Dynamic, Open Source Programming Language*. Python Software Foundation. Version 3.13.
77. Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, et al. 2019. “PyTorch: An Imperative Style, High-Performance Deep Learning Library.” In *Advances in Neural Information Processing Systems* 32, 8024–8035. Curran Associates. Preprint: arXiv:1912.01703.
78. Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. “Array Programming with NumPy.” *Nature* 585, no. 7825 (2020): 357–362.
79. Wes McKinney 2010. “Data Structures for Statistical Computing in Python.” In *Proceedings of the 9th Python in Science Conference*, 56–61.
80. Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python.” *Nature Methods* 17, no. 3 (2020): 261–272.
81. John D. Hunter. “Matplotlib: A 2D Graphics Environment.” *Computing in Science & Engineering* 9, no. 3 (2007): 90–95.

FIGURE LEGENDS

- Figure 1.** One realisation of each data-generating process at $n = 96$. The context supplied to every method is in grey and the held-out horizon in colour, separated by the dotted line.
- Figure 2.** Accuracy of TimesFM-3 relative to the best classical method in each scenario, on a $\log_2$ scale. Bars below zero mark scenarios where TimesFM-3 is more accurate. The label under each process names the family correctly specified for it by construction.
- Figure 3.** Pairwise comparisons of TimesFM-3 with each classical method, scenario by scenario: median of the paired per-series ratios of MASE over $h = 1, \dots, 12$, with 95% bootstrap intervals, on a logarithmic scale. Filled markers are significant after Benjamini–Hochberg correction within each opponent family; points left of 1 favour TimesFM-3. Panel titles give the significant comparisons won and lost by TimesFM-3. Arrowheads mark medians beyond the plotted range.
- Figure 4.** Empirical coverage of the nominal 60% and 80% intervals, averaged over all nine processes, against series length. The nominal level is dashed. TimesFM-3’s 80% coverage stays within 0.802–0.834 across every length; the classical methods under-cover at $n = 24$ and reach or exceed nominal from $n = 96$.
- Figure 5.** Typical against worst-case accuracy of the fifteen methods of the main design: geometric mean over the 36 scenarios of each method’s ratio to the best method in a scenario, and the largest such ratio, with 95% bootstrap intervals. Circles are foundation models, squares classical methods; lower and further left is better on both counts.
- Figure 6.** Simulated and M4 Monthly series in a four-feature instance space (spectral entropy, trend strength, seasonal strength and first-order autocorrelation of the differences), first two principal components. Grey points are M4 series; coloured points are simulated series at $n = 96$. (a) The fixed-parameter design covers 48.5% of the M4 series; (b) the randomised-parameter design covers 75.8%, with coverage defined in Section 6.1.